

Shen Zhen MTC Co., LTD

TEST REPORT

SCOPE OF WORK

EMC TESTING—MUDV50B0-90106-T2, MUDV50B0-90106-S2,
MUDV50B0-90106, MUDV50****-90106-## (*, # can from 0 to 9, A
to Z or blank), 4T-C50FJ2KL1FB, 4T-C50FJ2EL2NB, 4T-C50HJ4020EB
4T-C50HJ4020KB, **_50***** (* can from 0 to 9, A to Z or
blank), PR50U3421SKB

REPORT NUMBER

260508031SZN-001

ISSUE DATE

21 May 2026 [-----]

[REVISED DATE]

PAGES

55

DOCUMENT CONTROL NUMBER

EN55032/35_MMEa

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EMC VERIFICATION SUMMARY

Report No.: 260508031SZN-001

☒ATV ☒DVB-T/T2 ☒DVB-C ☒AV In ☒USB Play ☒HDMI In ☒HDMI (ARC)☒IPTV

Model: MUDV50B0-90106-T2 MUDV50B0-90106-S2, MUDV50B0-90106, MUDV50****-90106-## (*, # can from 0 to 9, A to Z or blank), 4T-C50FJ2KL1FB, 4T-C50FJ2EL2NB, 4T-C50HJ4020EB, 4T-C50HJ4020KB, **-*50*****(* can from 0 to 9, A to Z or blank), PR50U3421SKB. Product Description: TV/LCD LED TV/LED LCD TV Brand names: AMTC, Sharp, Polaroid	Applicant: Shen Zhen MTC Co., LTD MTC Industry Park, 1st Lilang Road, Xialilang community, Nanwan Street, Longgang district, Shenzhen, China Sample Receipt Date: 01 December 2022, 31 July 2024 Test Conducted Date: 01 December 2022 to 06 January 2023, 31 July 2024 to 15 August 2024		
☒ 1 st TEST ☐ 2 nd TEST	ALL TESTS WERE CONDUCTED IN ACCORDANCE WITH: *EN 55032: 2015+A11:2020 *EN IEC61000-3-2: 2019+A1:2021 *EN 61000-3-3: 2013+A2:2021 *EN 55035: 2017+A11:2020		
Test Site and Location:	Intertek Testing Services Shenzhen Ltd. No. 101&201, Building B, No. 308 Wuhe Avenue, Zhangkengjing Community, GuanHu Subdistrict, LongHua District, ShenZhen, P.R. China.		
Test Result	Pass	Fail	See Remark
*EN 55032: 2015+A11:2020	☒	☐	☐
*EN IEC61000-3-2: 2019+A1:2021	☒	☐	☐
*EN 61000-3-3: 2013+A2:2021	☒	☐	☐
*EN 55035: 2017+A11:2020	☒	☐	☐
When determining the test conclusion, the Measurement Uncertainty of test has been considered. Note: The highest frequency of the internal sources of the EUT is 5.8 GHz, the measurement shall be made up to 6 GHz.			

Prepared and Checked By:

Approved By:

Signature

Bruce Zheng
Project Engineer
21 May 2026

Date

Signature

Rode Liu
Project Engineer
21 May 2026

Date

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EMC Results Conclusion (with Justification)

RE: EMC Testing Pursuant to EMC Directive 2014/30/EU Performed On The TV/LCD LED TV/LED LCD TV,

Model: MUDV50B0-90106-T2

MUDV50B0-90106-S2, MUDV50B0-90106, MUDV50****-90106-## (*, # can from 0 to 9, A to Z or blank), 4T-C50FJ2KL1FB, 4T-C50FJ2EL2NB, 4T-C50HJ4020EB, 4T-C50HJ4020KB, **-*50*****(* can from 0 to 9, A to Z or blank), PR50U3421SKB.

We tested the TV/LCD LED TV/LED LCD TV, Model: MUDV50B0-90106-T2, to determine if it was in compliance with the relevant EN standards as marked on the EMC Verification Summary. We found that the unit met the requirement of EN 55032, EN IEC 61000-3-2, EN 61000-3-3, EN 55035 standards when tested as received.

The Model: MUDV50B0-90106-S2, MUDV50B0-90106, MUDV50****-90106-## (*, # can from 0 to 9, A to Z or blank), 4T-C50FJ2KL1FB, 4T-C50FJ2EL2NB, 4T-C50HJ4020EB, 4T-C50HJ4020KB, **-*50*****(* can from 0 to 9, A to Z or blank), PR50U3421SKB are the same as the Model: MUDV50B0-90106-T2 in hardware aspect. The difference in model number and Trade name serves as marketing strategy.

This report is based on previous report with report number 240731080SZN-001 dated 10 September 2024(original signatory: Bruce Zheng, Rode Liu on file). Owing to adding the models, no test was required after evaluation.

The production units are required to conform to the initial sample as received when the units are placed on the market.

LABORATORY MEASUREMENTS

Configuration Information

Equipment Under Test (EUT):	TV/LCD LED TV/LED LCD TV
Model:	MUDV50B0-90106-T2
Serial No.:	N/A
Support Equipment:	TV Signal Generator (Fluke 54200M01) DTV Multisignal Generator(EIDEN MSD5000A) USB Memory (TOSHIBA UHYBS-004G-BL) Laptop (Lenovo T420) (Provided by Intertek)
Cables:	HDMI Cable: Shield, 120cm*2 AV Cable: un-shield, 120cm Network cable, un-shield, 120cm (Provided by Intertek)
Adaptor:	N/A
Rated Voltage:	220-240V~, 50/60Hz, 128W
Test System:	PAL BG

Performance Criteria for Immunity

The performance criteria are referred to the test standard: EN 55035

Performance criteria A

The equipment shall continue to operate as intended without operator intervention. No degradation of performance, loss of function or change of operating state is allowed below a performance level specified by the manufacturer when the equipment is used as intended. The performance level may be replaced by a permissible loss of performance. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and by what the user may reasonably expect from the equipment if used as intended.

Evaluation of Audio Quality

The measured acoustic interference ratio and/or the measured electrical interference ratio during the test shall be –20 dB or better.

Evaluation of Video Quality

In the evaluation of picture interference the wanted test signal produces a standard picture (in the case of video tape equipment on the screen of the test-TV-set) and the unwanted signal produces a degradation of the picture. The degradation may be in a number of forms, such as a superposed pattern, positional disturbance of synchronization, geometrical distortion, loss of picture contrast or brightness, artefacts, freezing or disturbance of motion, image loss, video data or decoding errors, etc.

Performance criteria B

During the application of the disturbance, degradation of performance is allowed. However, no unintended change of actual operating state or stored data is allowed to persist after the test.

After the test, the equipment shall continue to operate as intended without operator intervention; no degradation of performance or loss of function is allowed, below a performance level specified by the manufacturer, when the equipment is used as intended. The performance level may be replaced by a permissible loss of performance.

If the minimum performance level (or the permissible performance loss), or recovery time, is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and by what the user may reasonably expect from the equipment if used as intended.

Performance criteria C

Loss of function is allowed, provided the function is self-recoverable, or can be restored by the operation of the controls by the user in accordance with the manufacturer's instructions. A reboot or re-start operation is allowed.

Information stored in non-volatile memory, or protected by a battery backup, shall not be lost.

DRAFT

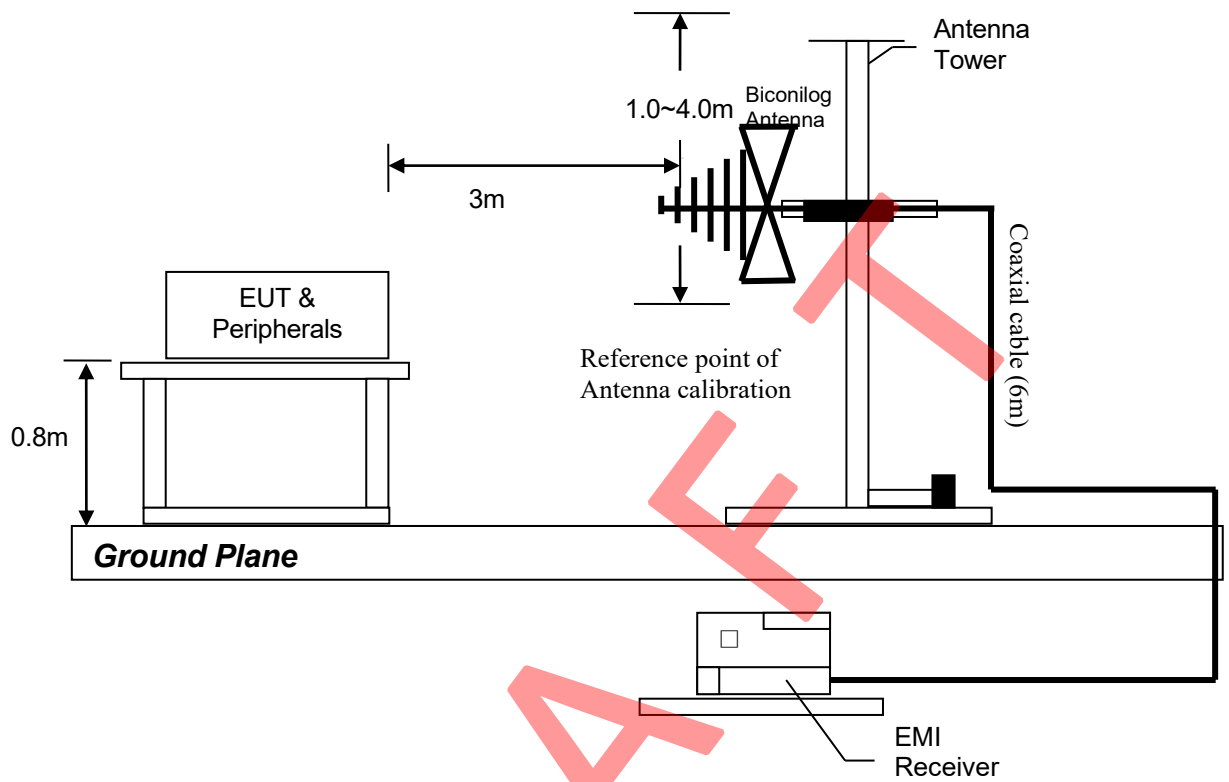
Radiated Disturbance
Pursuant to EN 55032: Emissions Requirement

Used Test Equipment

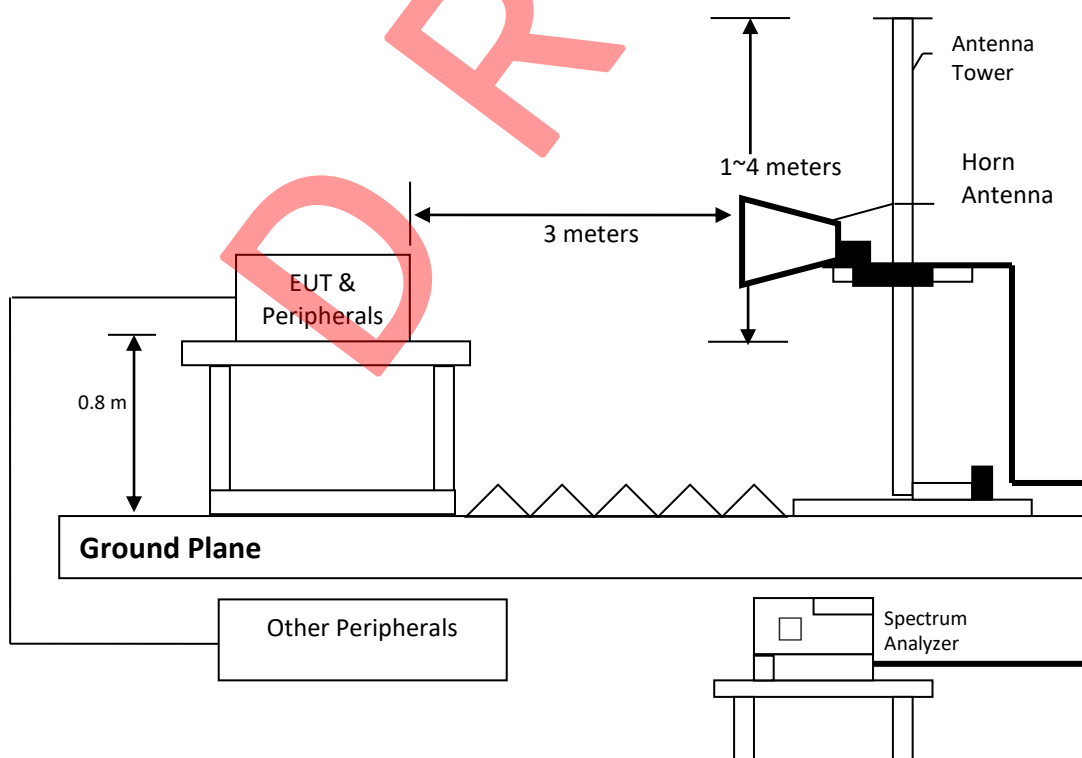
Equip No.	Description	Manufacturer	Model No.	Cal. Date	Due Date
SZ185-03	EMI Receiver	R & S	ESR7	2024-04-23	2025-04-23
SZ061-13	Biconilog Antenna	ETS	3142E	2022-07-13	2025-07-13
SZ061-08	Horn Antenna	ETS	3115	2021-09-05	2024-09-05
SZ056-03	Spectrum Analyzer	R & S	FSP30	2024-04-22	2025-04-22
SZ188-01	Anechoic Chamber	ETS	RFD-F/A-100	2021-12-12	2024-12-12
SZ181-04	Preamplifier	Agilent	8449B	2024-04-22	2025-04-22

- Notes:
1. Peak detector quick scan is showed on the graph and final quasi-peak detector data is measured corresponding to relevant limit and recorded in the data table. Only the worst-case data is recorded in the report.
 2. Frequency range scanned: 30MHz to 6000MHz.
 3. Only emissions significantly above equipment noise floor are reported.
 4. Uncertainty: 4.8dB at a level of confidence of 95%.

Test Setup Diagram:



(Radiated Emission Measurements Test Setup for 30MHz to 1GHz)



(Radiated Emission Measurements Test Setup for 1GHz to 6GHz)

Model: MUDV50B0-90106-T2

Intertek Report No.: 260508031SZN-001

Worst Case Operating Mode: DVB-T(506MHz)

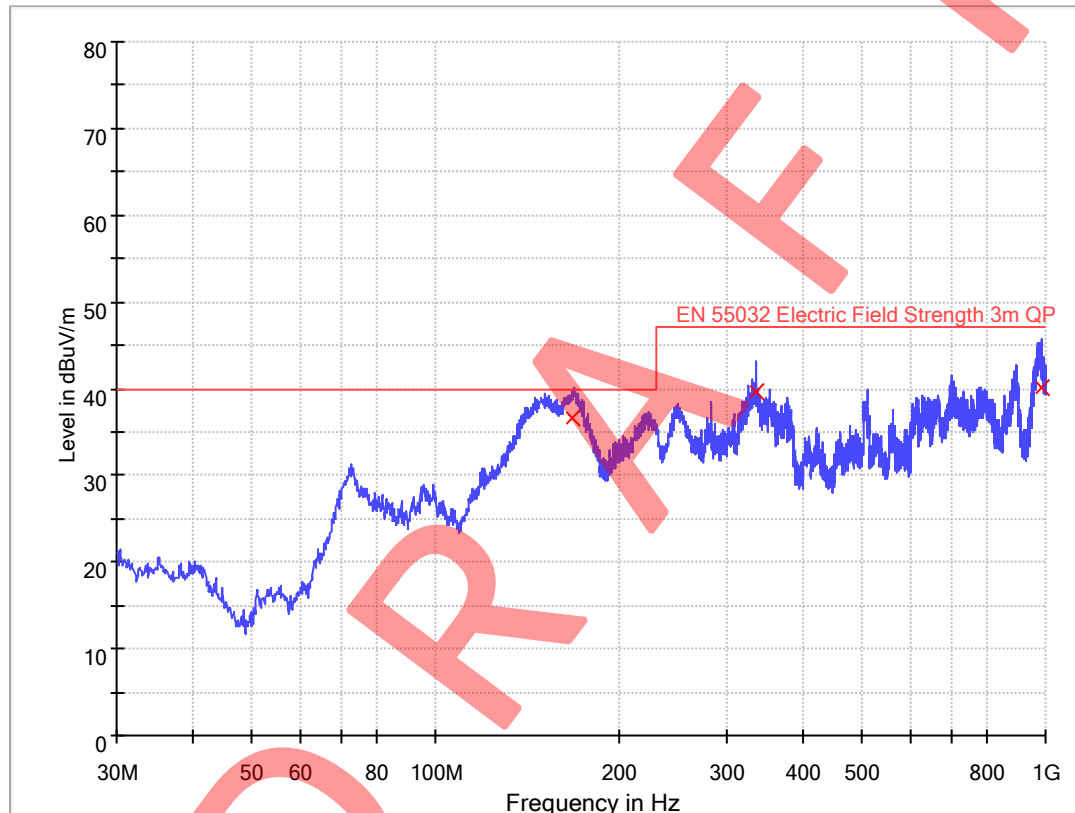
Worst Case Testing Voltage: AC 230V, 50Hz

Panel Model: V500D17-QE1

Test Data

Radiated Disturbance Pursuant to EN 55032: Emissions Requirement

Horizontal



Limit and Margin QP

Frequency (MHz)	QuasiPeak (dBuV/m)	Meas. Time (ms)	Bandwidth (kHz)	Polarization	Corr. (dB/m)	Margin - QPK (dB)	Limit - QPK (dBuV/m)
167.217350	36.6	1000.0	120.000	H	11.7	3.4	40.0
335.913750	39.7	1000.0	120.000	H	17.7	7.3	47.0
983.388750	40.0	1000.0	120.000	H	28.2	7.0	47.0

Remark:

1. Corr. (dB/m) = Antenna Factor (dB/m) + Cable Loss (dB)
2. QuasiPeak (dBuV/m) = Corr. (dB/m) + Read Level (dBuV)
3. Margin (dB) = Limit QPK (dBuV/m) - QuasiPeak (dBuV/m)

Model: MUDV50B0-90106-T2

Intertek Report No.: 260508031SZN-001

Worst Case Operating Mode: DVB-T(506MHz)

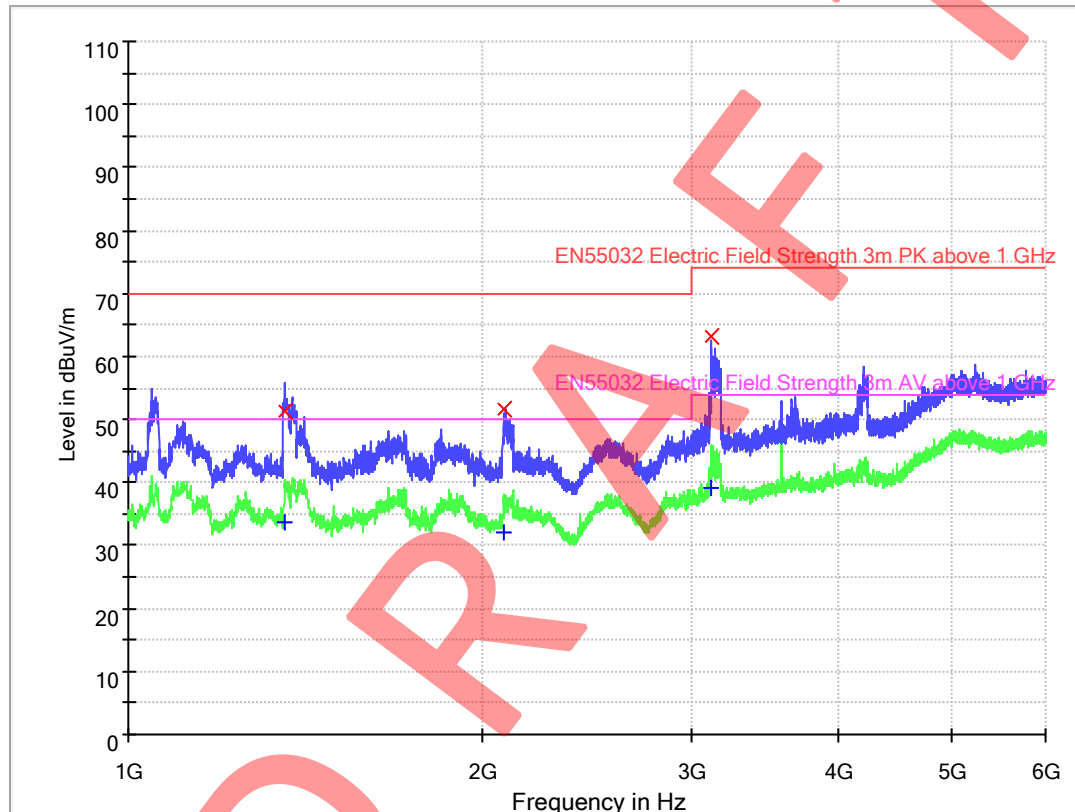
Worst Case Testing Voltage: AC 230V, 50Hz

Panel Model: V500D17-QE1

Test Data

Radiated Disturbance Pursuant to EN55032: Emissions Requirement

Horizontal



Limit and Margin PK

Frequency (MHz)	MaxPeak (dBuV/m)	Meas. Time (ms)	Bandwidth (kHz)	Polarization	Corr. (dB/m)	Margin - PK+ (dB)	Limit - PK+ (dBuV/m)
1356.000000	51.4	1000.0	1000.000	H	-7.1	18.6	70.0
2086.250000	51.6	1000.0	1000.000	H	-3.9	18.4	70.0
3125.625000	63.2	1000.0	1000.000	H	1.9	10.8	74.0

Limit and Margin AV

Frequency (MHz)	Average (dBuV/m)	Meas. Time (ms)	Bandwidth (kHz)	Polarization	Corr. (dB/m)	Margin - AVG (dB)	Limit - AVG (dBuV/m)
1356.000000	33.6	1000.0	1000.000	H	-7.1	16.4	50.0
2086.250000	32.0	1000.0	1000.000	H	-3.9	18.0	50.0
3125.625000	39.2	1000.0	1000.000	H	1.9	14.8	54.0

Remark:

1. Corr. (dB/m) = Antenna Factor (dB/m) + Cable Loss (dB) - Amp. Gain (dB)
2. Emission (dBuV/m) = Corr. (dB/m) + Read Level (dBuV)
3. Margin (dB) = Limit (dBuV) - Max peak/Average (dBuV)

Model: MUDV50B0-90106-T2

Intertek Report No.: 260508031SZN-001

Worst Case Operating Mode: DVB-T(506MHz)

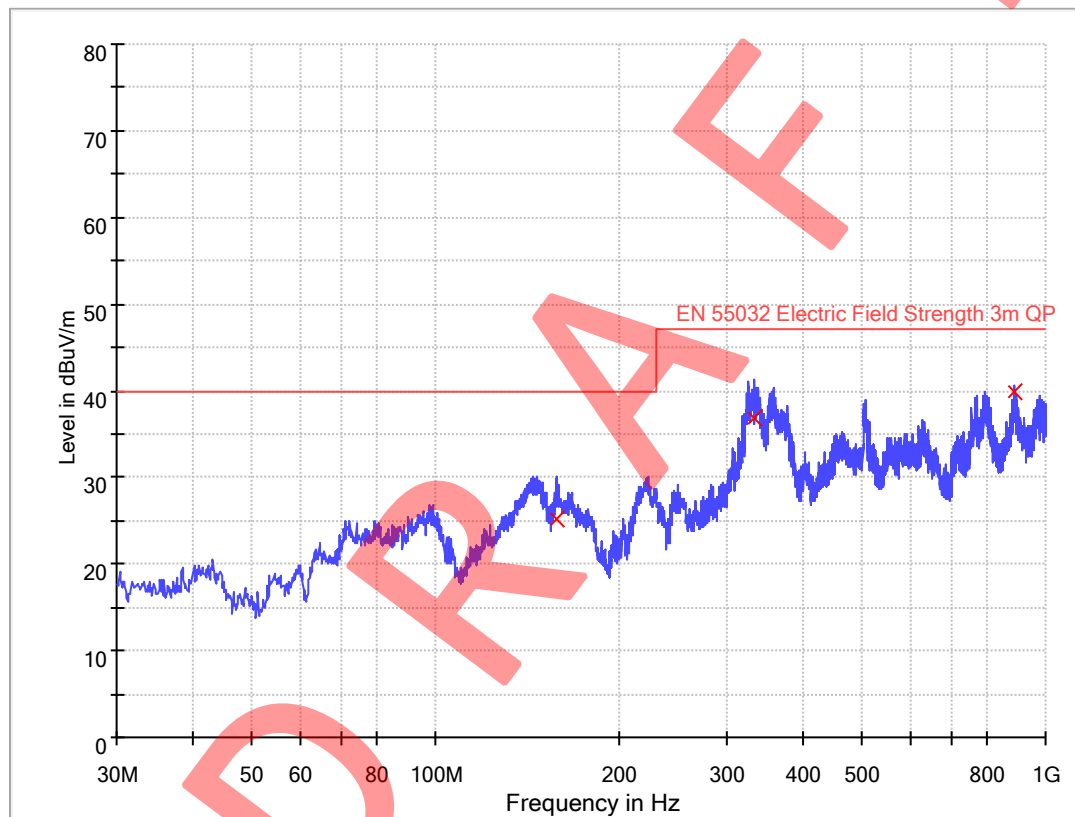
Worst Case Testing Voltage: AC 230V, 50Hz

Panel Model: V500D17-QE1

Test Data

Radiated Disturbance Pursuant to EN 55032: Emissions Requirement

Vertical



Limit and Margin QP

Frequency (MHz)	QuasiPeak (dBuV/m)	Meas. Time (ms)	Bandwidth (kHz)	Polarization	Corr. (dB/m)	Margin - QPK (dB)	Limit - QPK (dBuV/m)
157.676250	25.1	1000.0	120.000	V	12.0	14.9	40.0
332.120000	36.9	1000.0	120.000	V	17.5	10.1	47.0
888.935000	39.8	1000.0	120.000	V	27.3	7.2	47.0

Remark:

1. Corr. (dB/m) = Antenna Factor (dB/m) + Cable Loss (dB)

2. QuasiPeak (dBuV/m) = Corr. (dB/m) + Read Level (dBuV)

3. Margin (dB) = Limit QPK(dBuV/m) – QuasiPeak (dBuV/m)

Model: MUDV50B0-90106-T2

Intertek Report No.: 260508031SZN-001

Worst Case Operating Mode: DVB-T(506MHz)

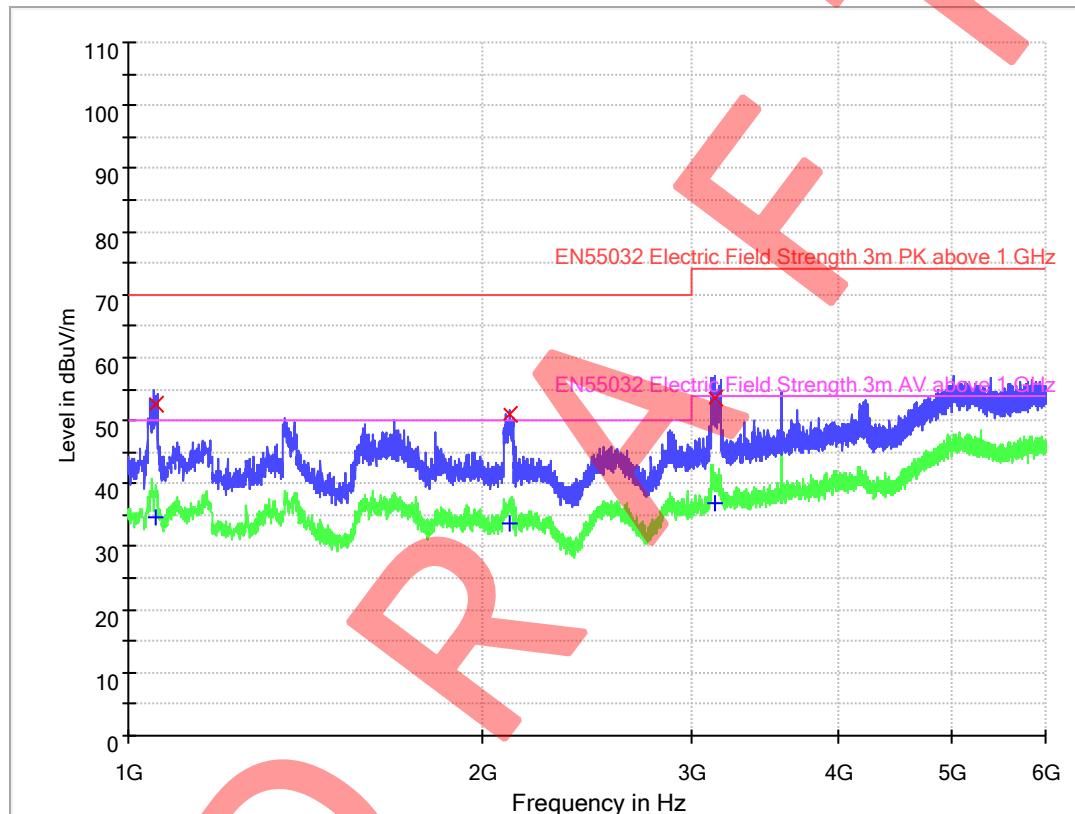
Worst Case Testing Voltage: AC 230V, 50Hz

Panel Model: V500D17-QE1

Test Data

Radiated Disturbance Pursuant to EN55032: Emissions Requirement

Vertical



Limit and Margin PK

Frequency (MHz)	MaxPeak (dBuV/m)	Meas. Time (ms)	Bandwidth (kHz)	Polarization	Corr. (dB/m)	Margin - PK+ (dB)	Limit - PK+ (dBuV/m)
1055.000000	54.1	1000.0	1000.000	V	-7.4	15.9	70.0
1355.000000	52.7	1000.0	1000.000	V	-7.2	17.3	70.0
3144.375000	54.6	1000.0	1000.000	V	2.0	19.4	74.0

Limit and Margin AV

Frequency (MHz)	Average (dBuV/m)	Meas. Time (ms)	Bandwidth (kHz)	Polarization	Corr. (dB/m)	Margin - AVG (dB)	Limit - AVG (dBuV/m)
1055.000000	34.0	1000.0	1000.000	V	-7.4	16.0	50.0
1355.000000	31.4	1000.0	1000.000	V	-7.2	18.6	50.0
3144.375000	36.4	1000.0	1000.000	V	2.0	17.6	54.0

Remark:

1. Corr. (dB/m) = Antenna Factor (dB/m) + Cable Loss (dB) - Amp. Gain (dB)
2. Emission (dBμV/m) = Corr. (dB/m) + Read Level (dBμV)
3. Margin (dB) = Limit (dBμV) - Max peak/Average (dBμV)

Model: MUDV50B0-90106-T2

Intertek Report No.: 260508031SZN-001

Worst Case Operating Mode: DVB-T(506MHz)

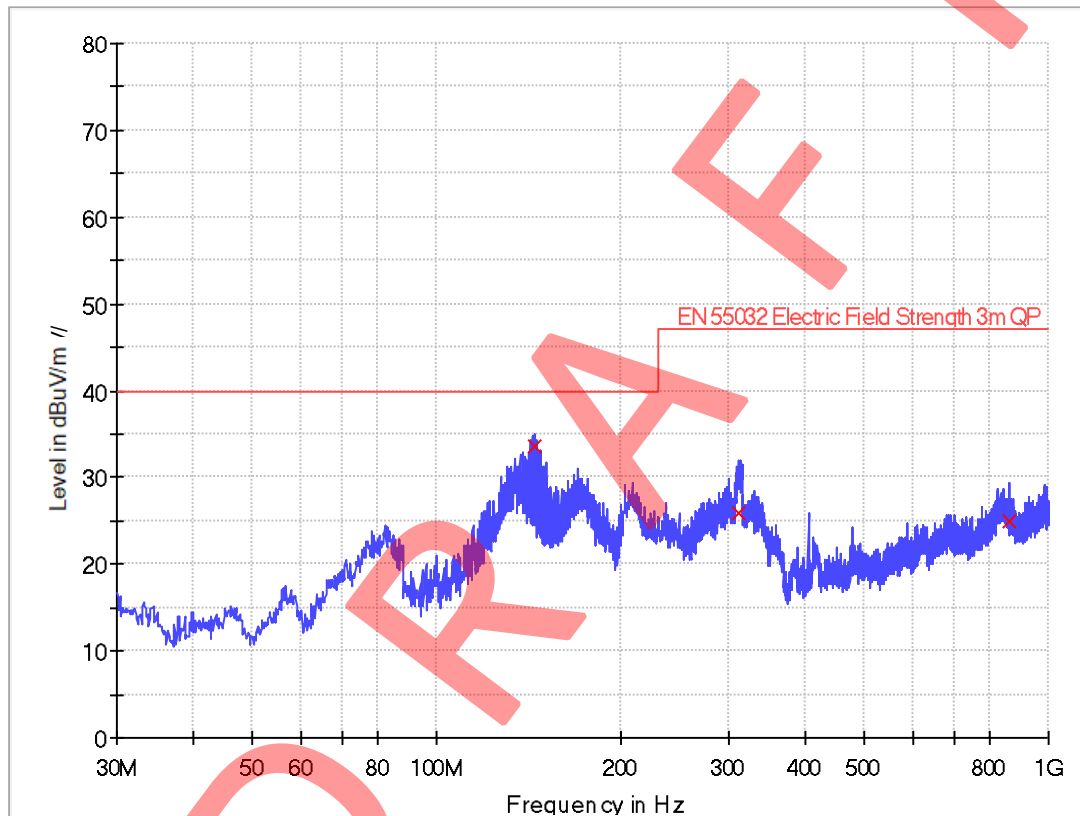
Worst Case Testing Voltage: AC 230V, 50Hz

Panel Model: HF500QUB

Test Data

Radiated Disturbance Pursuant to EN 55032: Emissions Requirement

Horizontal



Limit and Margin QP

Frequency (MHz)	QuasiPeak (dBuV/m)	Meas. Time (ms)	Bandwidth (kHz)	Polarization	Corr. (dB/m)	Margin - QPK (dB)	Limit - QPK (dBuV/m)
144.217500	33.5	1000.0	120.000	H	9.5	6.5	40.0
312.027500	25.8	1000.0	120.000	H	15.5	21.2	47.0
860.562500	25.0	1000.0	120.000	H	25.0	22.0	47.0

Remark:

1. Corr. (dB/m) = Antenna Factor (dB/m) + Cable Loss (dB)
2. QuasiPeak (dBuV/m) = Corr. (dB/m) + Read Level (dBuV)
3. Margin (dB) = Limit QPK (dBuV/m) - QuasiPeak (dBuV/m)

Model: MUDV50B0-90106-T2

Intertek Report No.: 260508031SZN-001

Worst Case Operating Mode: DVB-T(506MHz)

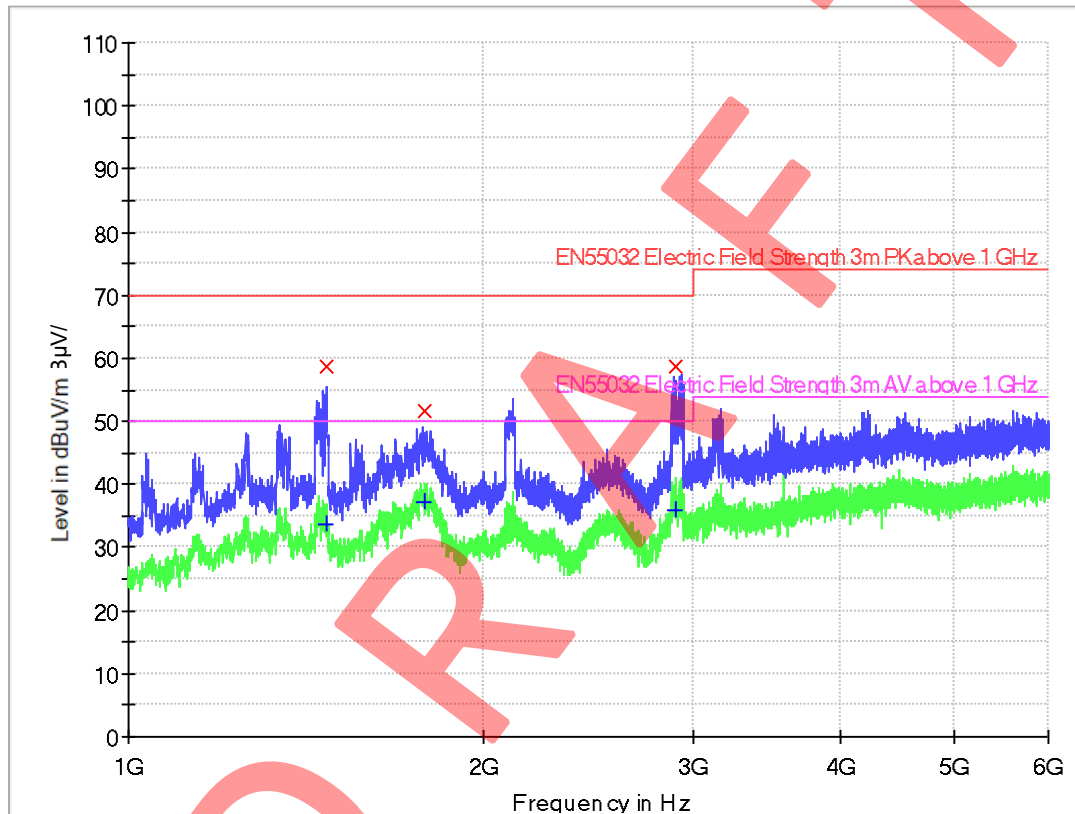
Worst Case Testing Voltage: AC 230V, 50Hz

Panel Model: HF500QUB

Test Data

Radiated Disturbance Pursuant to EN55032: Emissions Requirement

Horizontal



Limit and Margin PK

Frequency (MHz)	MaxPeak (dBuV/m)	Meas. Time (ms)	Bandwidth (kHz)	Polarization	Corr. (dB/m)	Margin - PK+ (dB)	Limit - PK+ (dBuV/m)
1470.625000	58.8	1000.0	1000.000	H	-7.2	11.2	70.0
1782.500000	51.5	1000.0	1000.000	H	-6.2	18.5	70.0
2903.750000	58.8	1000.0	1000.000	H	0.5	11.2	70.0

Limit and Margin AV

Frequency (MHz)	Average (dBuV/m)	Meas. Time (ms)	Bandwidth (kHz)	Polarization	Corr. (dB/m)	Margin - AVG (dB)	Limit - AVG (dBuV/m)
1470.625000	33.7	1000.0	1000.000	H	-7.2	16.3	50.0
1782.500000	37.2	1000.0	1000.000	H	-6.2	12.8	50.0
2903.750000	35.8	1000.0	1000.000	H	0.5	14.2	50.0

Remark:

1. Corr. (dB/m) = Antenna Factor (dB/m) + Cable Loss (dB) - Amp. Gain (dB)
2. Emission (dB μ V/m) = Corr. (dB/m) + Read Level (dB μ V)
3. Margin (dB) = Limit (dB μ V) - Max peak/Average (dB μ V)

Model: MUDV50B0-90106-T2

Intertek Report No.: 260508031SZN-001

Worst Case Operating Mode: DVB-T(506MHz)

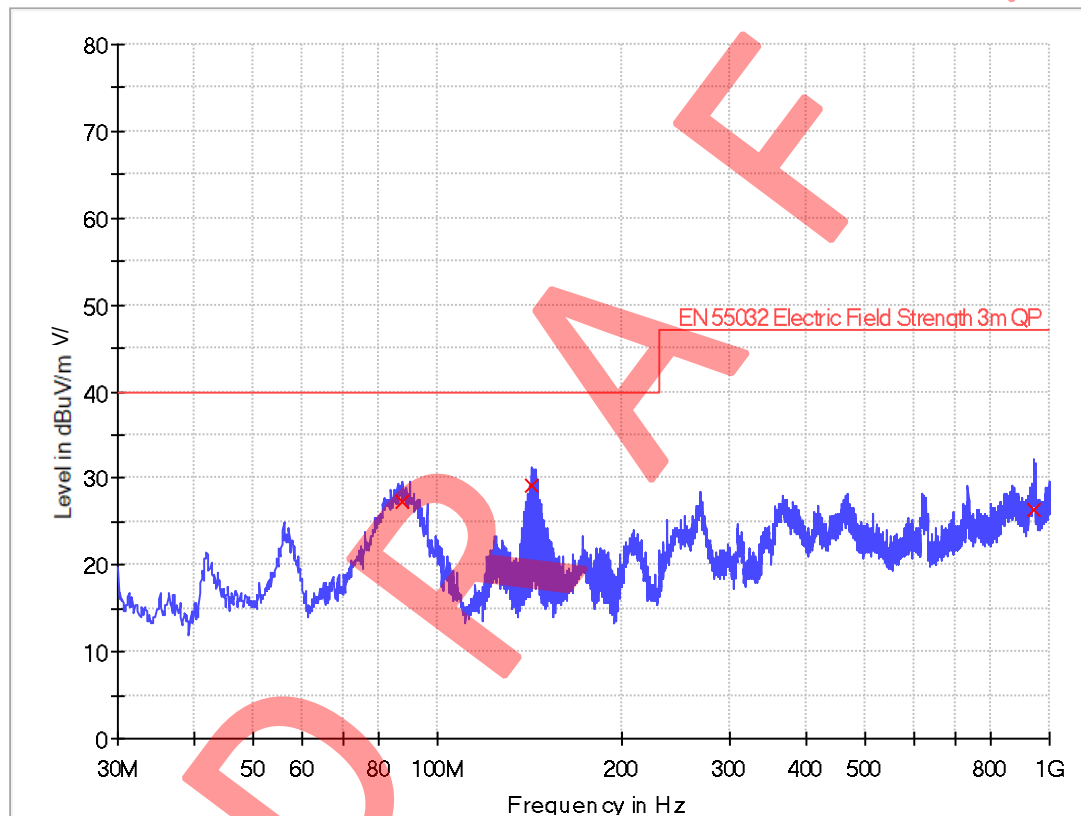
Worst Case Testing Voltage: AC 230V, 50Hz

Panel Model: HF500QUB

Test Data

Radiated Disturbance Pursuant to EN 55032: Emissions Requirement

Vertical



Limit and Margin QP

Frequency (MHz)	QuasiPeak (dBuV/m)	Meas. Time (ms)	Bandwidth (kHz)	Polarization	Corr. (dB/m)	Margin - QPK (dB)	Limit - QPK (dBuV/m)
87.593750	27.2	1000.0	120.000	V	8.6	12.8	40.0
142.762500	29.3	1000.0	120.000	V	9.4	10.7	40.0
942.770000	26.4	1000.0	120.000	V	25.7	20.6	47.0

Remark:

1. Corr. (dB/m) = Antenna Factor (dB/m) + Cable Loss (dB)
2. QuasiPeak (dBuV/m) = Corr. (dB/m) + Read Level (dBuV)
3. Margin (dB) = Limit QPK(dBuV/m) – QuasiPeak (dBuV/m)

Model: MUDV50B0-90106-T2

Intertek Report No.: 260508031SZN-001

Worst Case Operating Mode: DVB-T(506MHz)

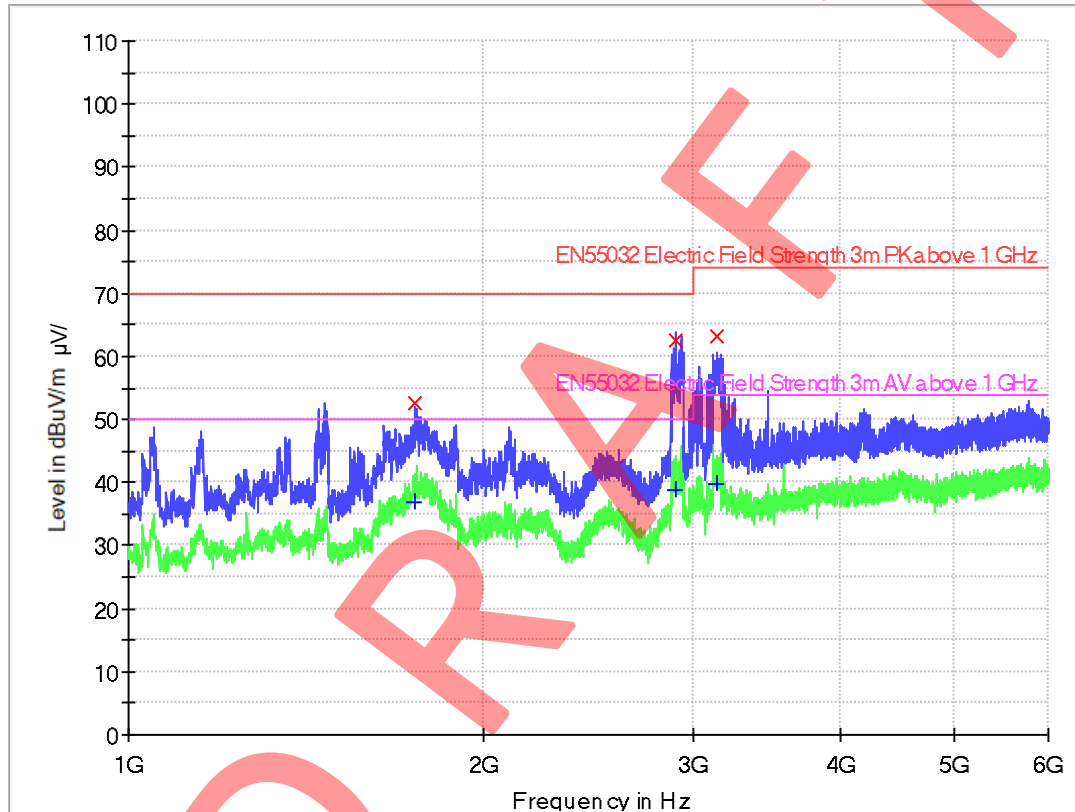
Worst Case Testing Voltage: AC 230V, 50Hz

Panel Model: HF500QUB

Test Data

Radiated Disturbance Pursuant to EN55032: Emissions Requirement

Vertical



Limit and Margin PK

Frequency (MHz)	MaxPeak (dBuV/m)	Meas. Time (ms)	Bandwidth (kHz)	Polarization	Corr. (dB/m)	Margin - PK+ (dB)	Limit - PK+ (dBuV/m)
1746.875000	52.6	1000.0	1000.000	V	-6.4	17.4	70.0
2907.500000	62.4	1000.0	1000.000	V	0.5	7.6	70.0
3140.625000	63.2	1000.0	1000.000	V	1.8	10.8	74.0

Limit and Margin AV

Frequency (MHz)	Average (dBuV/m)	Meas. Time (ms)	Bandwidth (kHz)	Polarization	Corr. (dB/m)	Margin - AVG (dB)	Limit - AVG (dBuV/m)
1746.875000	37.0	1000.0	1000.000	V	-6.4	13.0	50.0
2907.500000	38.7	1000.0	1000.000	V	0.5	11.3	50.0
3140.625000	39.6	1000.0	1000.000	V	1.8	14.4	54.0

Remark:

1. Corr. (dB/m) = Antenna Factor (dB/m) + Cable Loss (dB) - Amp. Gain (dB)
2. Emission (dBuV/m) = Corr. (dB/m) + Read Level (dBuV)
3. Margin (dB) = Limit (dBuV) - Max peak/Average (dBuV)

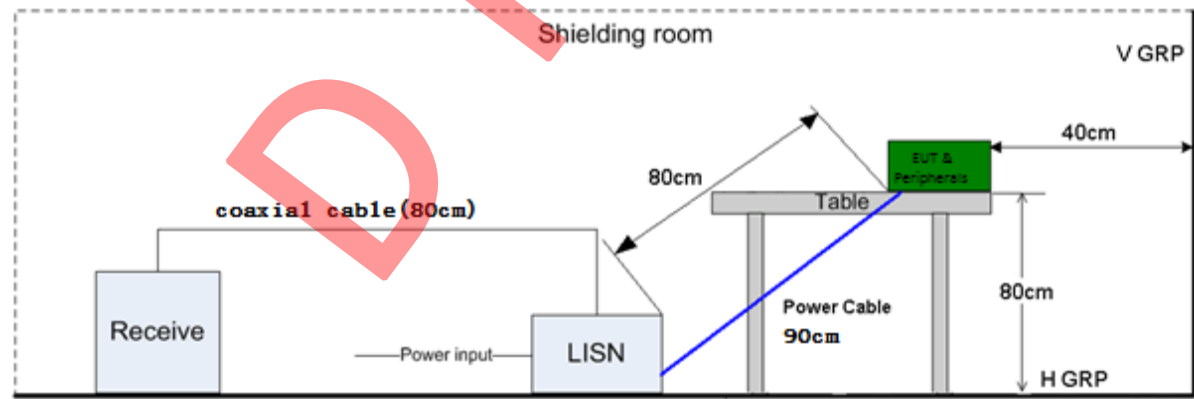
RFI Voltage Test
Pursuant to EN55032: Emissions Requirement

Used Test Equipment

Equip No.	Description	Manufacturer	Model No.	Cal. Date	Due Date
SZ185-02	EMI Receiver	R & S	ESCI	2024-07-09	2025-07-09
SZ187-01	Two-Line V-Network	R & S	ENV216	2023-10-18	2024-10-18
SZ187-02	Two-Line V-Network	R & S	ENV216	2024-04-23	2025-04-23
SZ188-03	Shielding Room	ETS	RFD-100	2022-12-20	2025-12-20

- Notes:
1. Peak and average detector quick scan are showed on the graph and final quasi-peak and average detector data are measured, the worst-case is recorded in the following graph and table.
 2. Frequency range scanned: 150kHz to 30MHz.
 3. Only emissions significantly above equipment noise floor is reported.
 4. Uncertainty: 3.2dB at a level of confidence of 95%.

Test Setup Diagram



Test set-up of conducted disturbance for Power port

Model: MUDV50B0-90106-T2

Intertek Report No.: 260508031SZN-001

Worst Case Operating Mode: DVB-T with antenna grounded

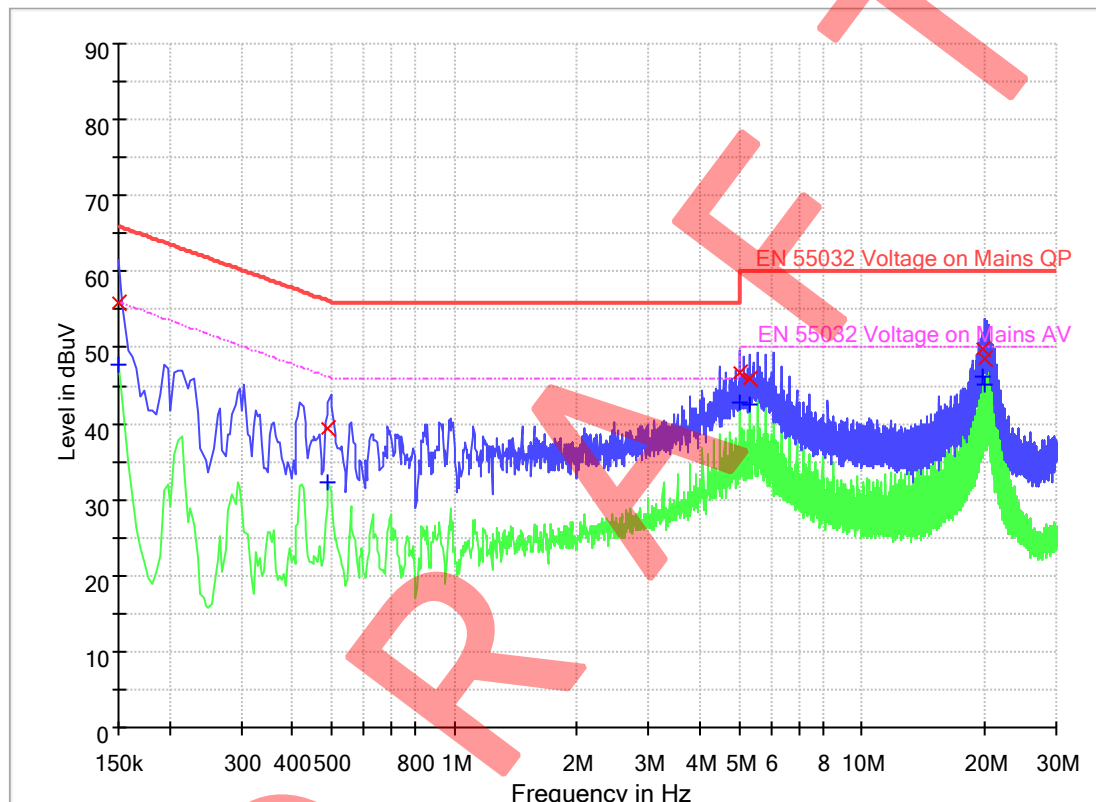
Phase: Live

Worst Case Testing Voltage: AC 230V, 60Hz

Panel Model: V500D17-QE1

Test Data

RFI Voltage Test Pursuant to EN 55032: Emissions Requirement



Limit and Margin QP

Frequency (MHz)	QuasiPeak (dBuV)	Bandwidth (kHz)	Line	Corr. (dB)	Margin (dB)	Limit (dBuV)
0.150000	55.8	9.000	L	9.6	10.2	66.0
0.490000	39.5	9.000	L	9.6	16.7	56.2
5.018000	46.8	9.000	L	9.8	13.2	60.0
5.286000	46.0	9.000	L	9.8	14.0	60.0
19.866000	50.0	9.000	L	10.4	10.0	60.0
19.998000	48.6	9.000	L	10.4	11.4	60.0

Limit and Margin AV

Frequency (MHz)	Average (dBuV)	Bandwidth (kHz)	Line	Corr. (dB)	Margin (dB)	Limit (dBuV)
0.150000	47.8	9.000	L	9.6	8.2	56.0
0.490000	32.4	9.000	L	9.6	13.8	46.2
5.018000	42.9	9.000	L	9.8	7.1	50.0
5.286000	42.5	9.000	L	9.8	7.5	50.0
19.866000	46.2	9.000	L	10.4	3.8	50.0
19.998000	45.0	9.000	L	10.4	5.0	50.0

Remark:

1. Corr. Factor (dB) = LISN Factor (dB) + Cable Loss (dB)
2. Margin (dB) = Limit (dBuV) – QuasiPeak/Average (dBuV)

Model: MUDV50B0-90106-T2

Intertek Report No.: 260508031SZN-001

Worst Case Operating Mode: DVB-T with antenna grounded

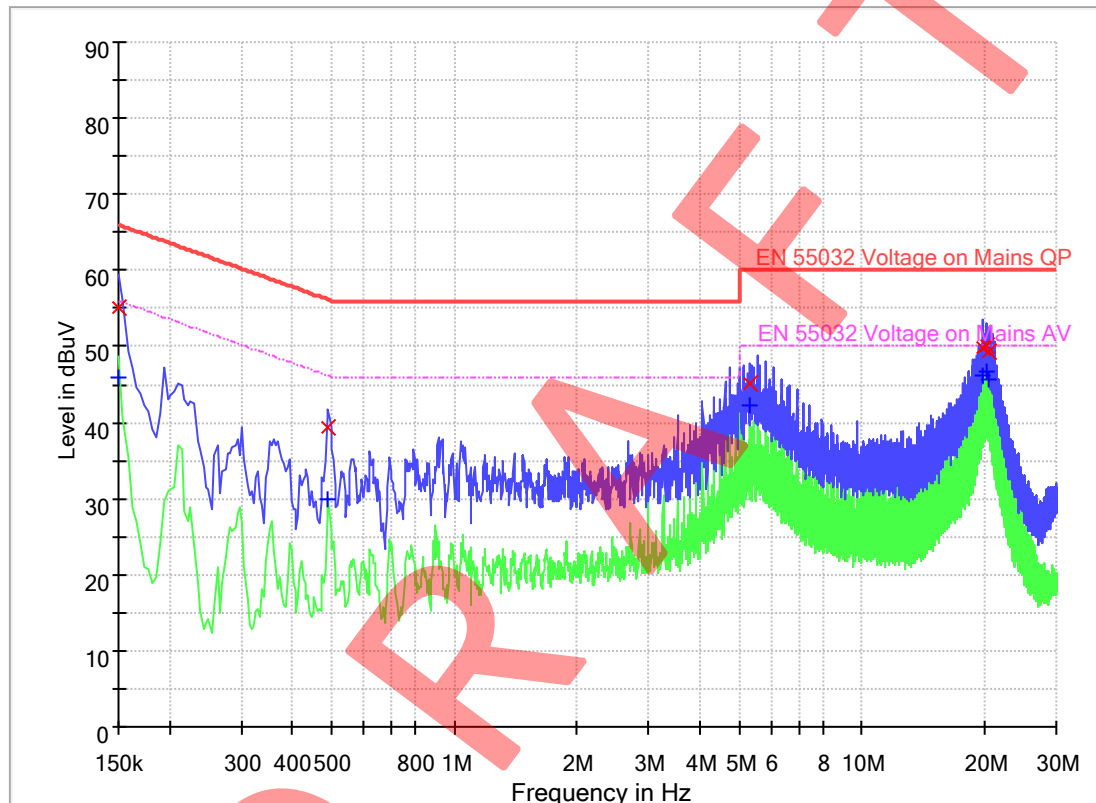
Phase: Neutral

Worst Case Testing Voltage: AC 230V, 50Hz

Panel Model: V500D17-QE1

Test Data

RFI Voltage Test Pursuant to EN 55032: Emissions Requirement



Limit and Margin QP

Frequency (MHz)	QuasiPeak (dBuV)	Bandwidth (kHz)	Line	Corr. (dB)	Margin (dB)	Limit (dBuV)
0.150000	55.1	9.000	N	9.6	10.9	66.0
0.490000	39.3	9.000	N	9.6	16.9	56.2
5.286000	45.1	9.000	N	9.8	14.9	60.0
19.866000	49.9	9.000	N	10.5	10.1	60.0
20.138000	50.2	9.000	N	10.5	9.8	60.0
20.410000	49.4	9.000	N	10.5	10.6	60.0

Limit and Margin AV

Frequency (MHz)	Average (dBuV)	Bandwidth (kHz)	Line	Corr. (dB)	Margin (dB)	Limit (dBuV)
0.150000	45.8	9.000	N	9.6	10.2	56.0
0.490000	29.8	9.000	N	9.6	16.4	46.2
5.286000	42.2	9.000	N	9.8	7.8	50.0
19.866000	46.2	9.000	N	10.5	3.8	50.0
20.138000	46.7	9.000	N	10.5	3.3	50.0
20.410000	45.6	9.000	N	10.5	4.4	50.0

Remark:

1. Corr. Factor (dB) = LISN Factor (dB) + Cable Loss (dB)
2. Margin (dB) = Limit (dBuV) – QuasiPeak/Average (dBuV)

Model: MUDV50B0-90106-T2

Intertek Report No.: 260508031SZN-001

Worst Case Operating Mode: HDMI in with antenna grounded

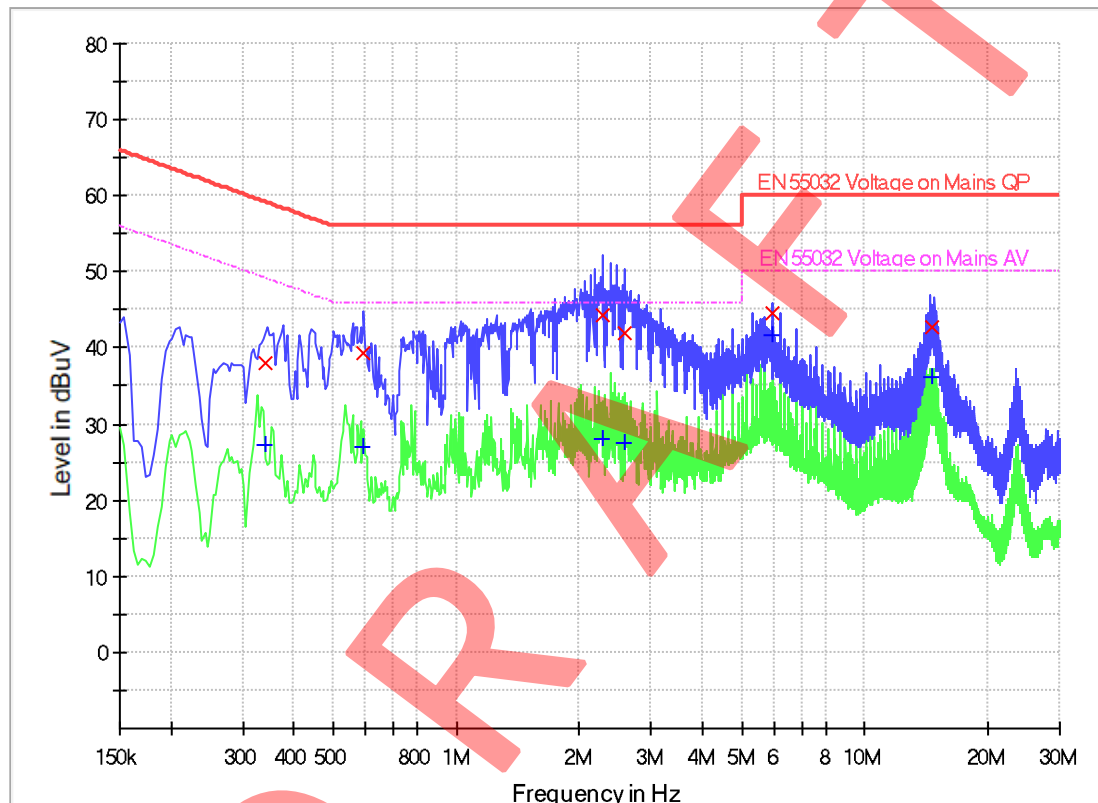
Phase: Live

Worst Case Testing Voltage: AC 230V, 60Hz

Panel Model:HF500QUB

Test Data

RFI Voltage Test Pursuant to EN 55032: Emissions Requirement



Limit and Margin QP

Frequency (MHz)	QuasiPeak (dBuV)	Bandwidth (kHz)	Line	Corr. (dB)	Margin (dB)	Limit (dBuV)
0.342000	38.1	9.000	L	9.6	21.1	59.2
0.594000	39.2	9.000	L	9.6	16.8	56.0
2.286000	44.4	9.000	L	9.7	11.6	56.0
2.582000	41.9	9.000	L	9.7	14.1	56.0
5.966000	44.6	9.000	L	9.8	15.4	60.0
14.578000	42.6	9.000	L	10.4	17.4	60.0

Limit and Margin AV

Frequency (MHz)	Average (dBuV)	Bandwidth (kHz)	Line	Corr. (dB)	Margin (dB)	Limit (dBuV)
0.342000	27.2	9.000	L	9.6	22.0	49.2
0.594000	27.0	9.000	L	9.6	19.0	46.0
2.286000	28.0	9.000	L	9.7	18.0	46.0
2.582000	27.4	9.000	L	9.7	18.6	46.0
5.966000	41.6	9.000	L	9.8	8.4	50.0
14.578000	36.2	9.000	L	10.4	13.8	50.0

Remark:

1. Corr. Factor (dB) = LISN Factor (dB) + Cable Loss (dB)
2. Margin (dB) = Limit (dBuV) – QuasiPeak/Average (dBuV)

Model: MUDV50B0-90106-T2

Intertek Report No.: 260508031SZN-001

Worst Case Operating Mode: HDMI in with antenna grounded

Phase: Neutral

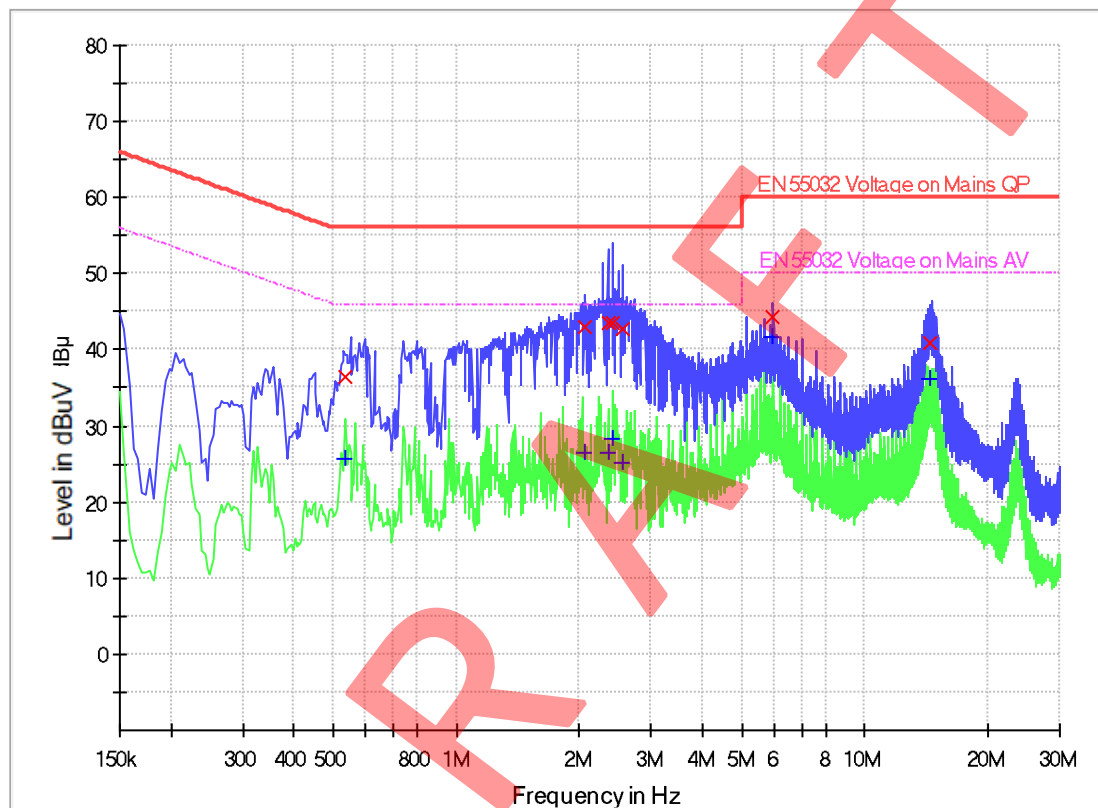
Worst Case Testing Voltage: AC 230V, 50Hz

Panel Model:HF500QUB

Test Data

RFI Voltage Test

Pursuant to EN 55032: Emissions Requirement



Limit and Margin QP

Frequency (MHz)	QuasiPeak (dBuV)	Bandwidth (kHz)	Line	Corr. (dB)	Margin (dB)	Limit (dBuV)
0.534000	36.3	9.000	N	9.6	19.7	56.0
2.070000	43.1	9.000	N	9.7	12.9	56.0
2.350000	43.5	9.000	N	9.7	12.5	56.0
2.406000	43.4	9.000	N	9.7	12.6	56.0
2.554000	42.8	9.000	N	9.7	13.2	56.0
5.966000	44.4	9.000	N	9.8	15.6	60.0
14.506000	41.0	9.000	N	10.4	19.0	60.0

Limit and Margin AV

Frequency (MHz)	Average (dBuV)	Bandwidth (kHz)	Line	Corr. (dB)	Margin (dB)	Limit (dBuV)
0.534000	25.7	9.000	N	9.6	20.3	46.0
2.070000	26.5	9.000	N	9.7	19.5	46.0
2.350000	26.4	9.000	N	9.7	19.6	46.0
2.406000	28.4	9.000	N	9.7	17.6	46.0
2.554000	25.3	9.000	N	9.7	20.7	46.0
5.966000	41.6	9.000	N	9.8	8.4	50.0
14.506000	36.2	9.000	N	10.4	13.8	50.0

Remark:

1. Corr. Factor (dB) = LISN Factor (dB) + Cable Loss (dB)
2. Margin (dB) = Limit (dBuV) – QuasiPeak/Average (dBuV)

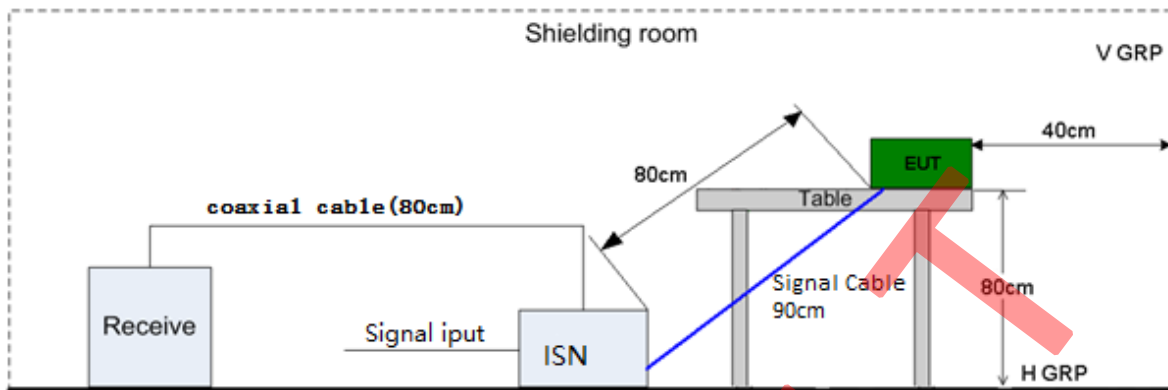
Asymmetric Mode Conducted Emissions
Pursuant to EN55032: Emissions Requirement

Used Test Equipment

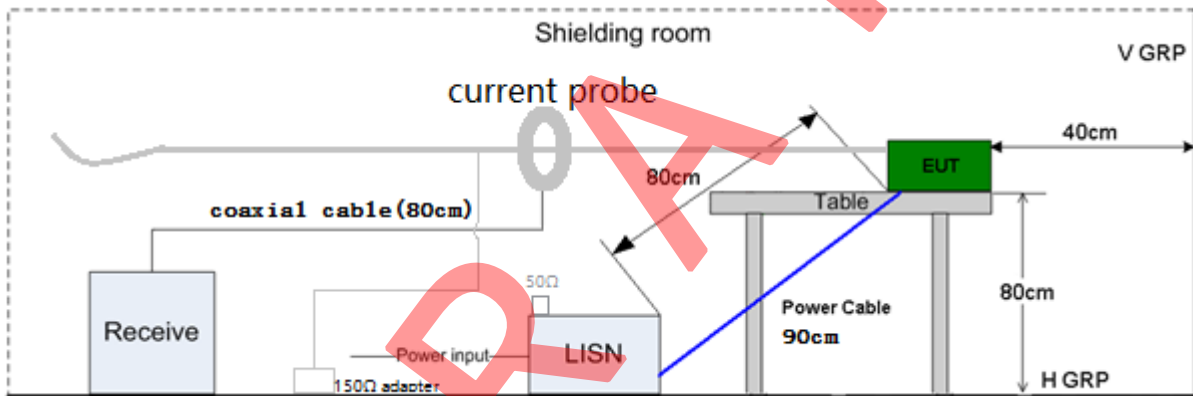
Equip No.	Description	Manufacturer	Model No.	Cal. Date	Due Date
SZ185-02	EMI Receiver	R & S	ESCI	2024-07-09	2025-07-09
SZ183-05	Current Transformer	SCHWARZBECK	SW 9605	2024-07-10	2025-07-10
SZ187-01	LISN	R & S	ENV216	2023-10-18	2024-10-18
SZ184-02	Impedance Stabilization Network	Teseq	ISN T4A	2023-12-12	2024-12-12
SZ188-03	Shielding Room	ETS	RFD-100	2022-12-20	2025-12-20

- Notes:
1. Peak and average detector quick scan are showed on the graph and final quasi-peak and average detector data are measured, the worst-case is recorded in the following graph and table.
 2. Frequency range scanned: 150 kHz to 30MHz.
 3. Only emissions significantly above equipment noise floor is reported.
 4. Uncertainty: $\pm 3.2\text{dB}$ at a level of confidence of 95%.

Test Setup Diagram



Test setup for Balanced Unscreened port



Test setup for Screened or Coaxial port

Model: MUDV50B0-90106-T2

Intertek Report No.: 260508031SZN-001

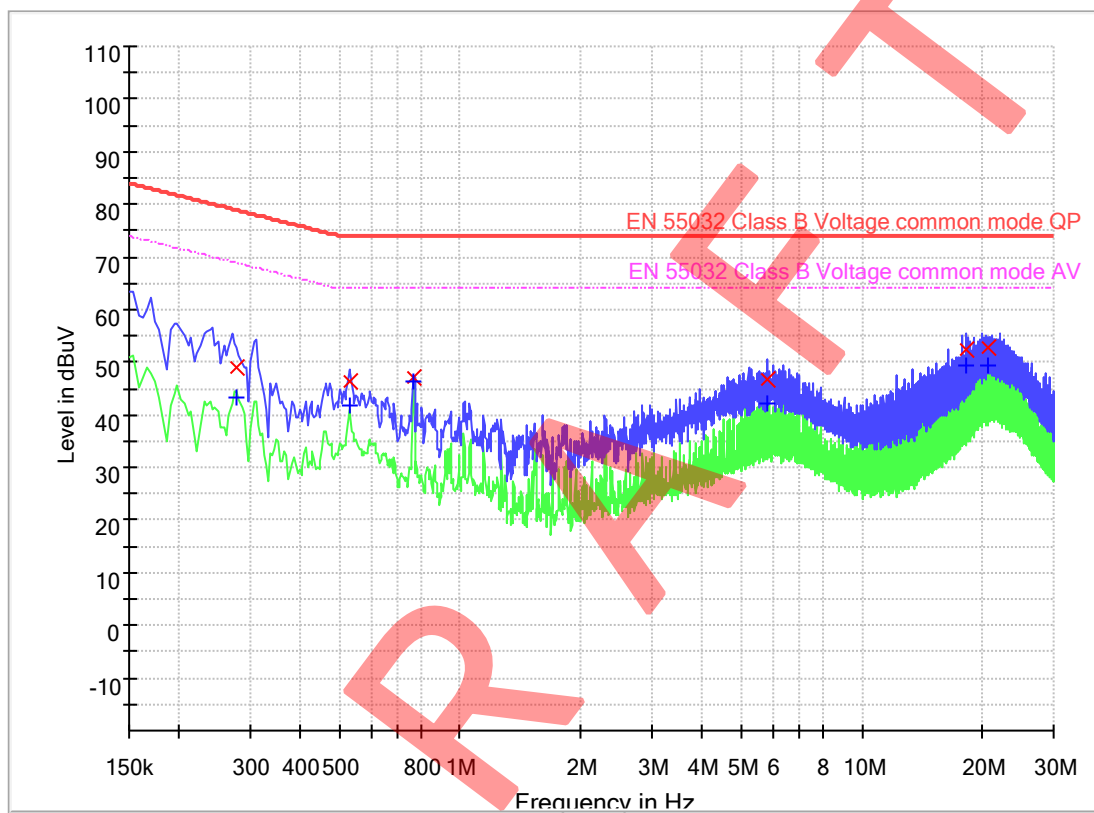
Worst Case Operating Mode: Ethernet

Port: Wired network port

Panel Model: V500D17-QE1

Test Data

Asymmetric Mode Conducted Emissions (LAN Port) Pursuant to EN55032: Emissions Requirement



Limit and Margin QP

Frequency (MHz)	QuasiPeak (dBuV)	Bandwidth (kHz)	Corr. (dB)	Margin (dB)	Limit (dBuV)
0.278000	48.9	9.000	9.7	30.0	78.9
0.530000	46.4	9.000	9.7	27.6	74.0
0.766000	47.1	9.000	9.7	26.9	74.0
5.830000	46.5	9.000	9.3	27.5	74.0
18.242000	52.3	9.000	9.4	21.7	74.0
20.542000	52.7	9.000	9.4	21.3	74.0

Limit and Margin AV

Frequency (MHz)	Average (dBuV)	Bandwidth (kHz)	Corr. (dB)	Margin (dB)	Limit (dBuV)
0.278000	43.5	9.000	9.7	25.4	68.9
0.530000	41.8	9.000	9.7	22.2	64.0
0.766000	46.5	9.000	9.7	17.5	64.0
5.830000	42.3	9.000	9.3	21.7	64.0
18.242000	49.5	9.000	9.4	14.5	64.0
20.542000	49.3	9.000	9.4	14.7	64.0

Remark:

1. Corr. Factor (dB) = ISN Factor (dB) + Cable Loss (dB)
2. Margin (dB) = Limit (dBuV) – QuasiPeak/Average (dBuV)

Model: MUDV50B0-90106-T2

Intertek Report No.: 260508031SZN-001

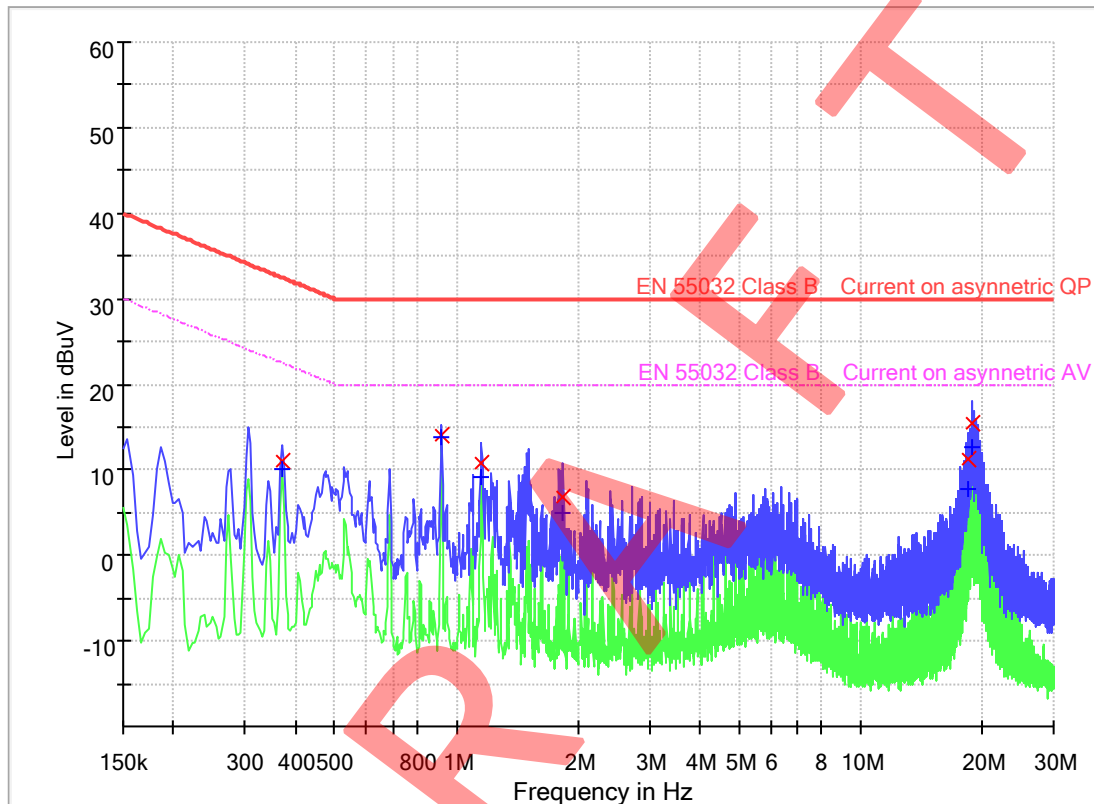
Worst Case Operating Mode: DVB-T(506MHz)

Port: DVB-T Antenna

Panel Model: V500D17-QE1

Test Data

Asymmetric Mode Conducted Emissions Pursuant to EN55032: Emissions Requirement



Limit and Margin QP

Frequency (MHz)	QuasiPeak (dBuA)	Bandwidth (kHz)	Corr. (dB)	Margin (dB)	Limit (dBuA)
0.370000	11.1	9.000	0.5	21.4	32.5
0.918000	14.1	9.000	0.2	15.9	30.0
1.156000	10.8	9.000	0.2	19.2	30.0
1.840000	6.8	9.000	0.2	23.2	30.0
18.512000	11.4	9.000	0.2	18.6	30.0
18.916000	15.5	9.000	0.2	14.5	30.0

Limit and Margin AV

Frequency (MHz)	Average (dBuA)	Bandwidth (kHz)	Corr. (dB)	Margin (dB)	Limit (dBuA)
0.370000	10.1	9.000	0.5	12.4	22.5
0.918000	13.9	9.000	0.2	6.1	20.0
1.156000	9.3	9.000	0.2	10.7	20.0
1.840000	4.9	9.000	0.2	15.1	20.0
18.512000	7.9	9.000	0.2	12.1	20.0
18.916000	12.6	9.000	0.2	7.4	20.0

Remark:

1. Corr. Factor (dB) = Current Probe Factor (dB) + Cable Loss (dB)

2. Margin (dB) = Limit (dBuV) – QuasiPeak/Average (dBuV)

Model: MUDV50B0-90106-T2

Intertek Report No.: 260508031SZN-001

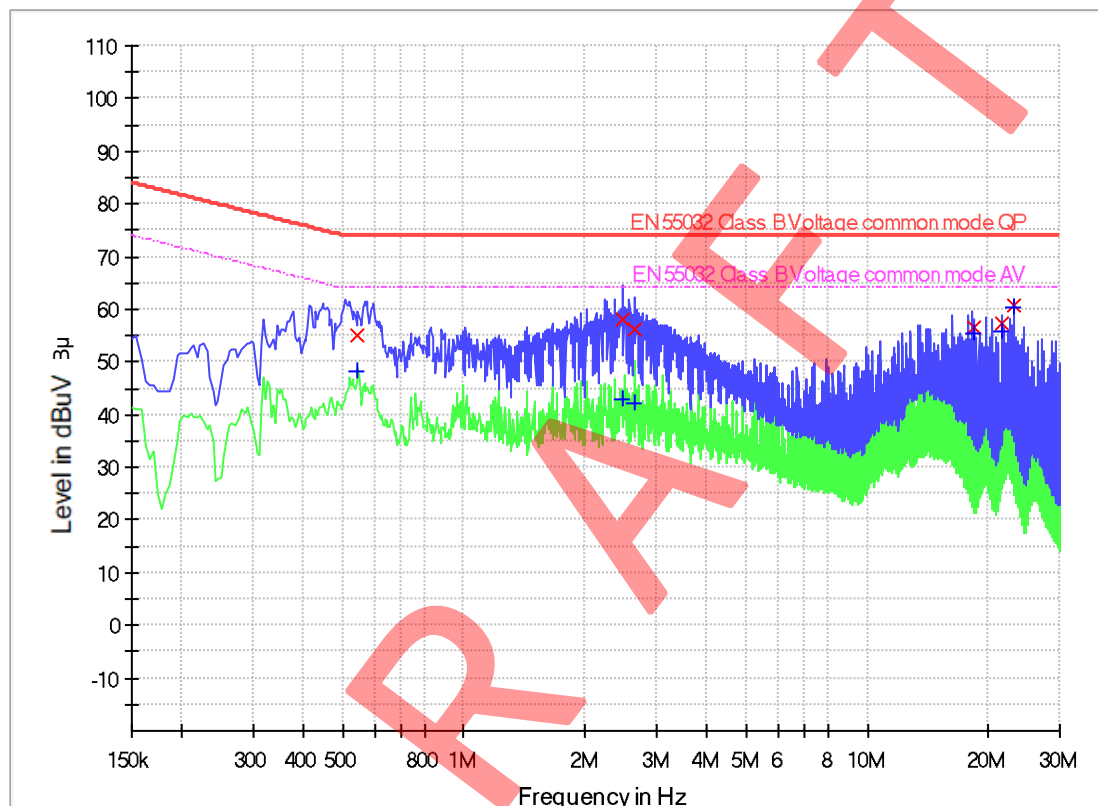
Worst Case Operating Mode: Ethernet

Port: Wired network port

Panel Model: HF500QUB

Test Data

Asymmetric Mode Conducted Emissions (LAN Port) Pursuant to EN55032: Emissions Requirement



Limit and Margin QP

Frequency (MHz)	QuasiPeak (dBuV)	Bandwidth (kHz)	Corr. (dB)	Margin (dB)	Limit (dBuV)
0.546000	55.0	9.000	9.7	19.0	74.0
2.478000	58.1	9.000	9.4	15.9	74.0
2.650000	56.1	9.000	9.4	17.9	74.0
18.306000	56.5	9.000	9.4	17.5	74.0
21.662000	57.3	9.000	9.4	16.7	74.0
23.130000	60.9	9.000	9.4	13.1	74.0

Limit and Margin AV

Frequency (MHz)	Average (dBuV)	Bandwidth (kHz)	Corr. (dB)	Margin (dB)	Limit (dBuV)
0.546000	48.1	9.000	9.7	15.9	64.0
2.478000	43.0	9.000	9.4	21.0	64.0
2.650000	42.2	9.000	9.4	21.8	64.0
18.306000	55.3	9.000	9.4	8.7	64.0
21.662000	55.8	9.000	9.4	8.2	64.0
23.130000	60.3	9.000	9.4	3.7	64.0

Remark:

1. Corr. Factor (dB) = ISN Factor (dB) + Cable Loss (dB)
2. Margin (dB) = Limit (dBuV) – QuasiPeak/Average (dBuV)

Model: MUDV50B0-90106-T2

Intertek Report No.: 260508031SZN-001

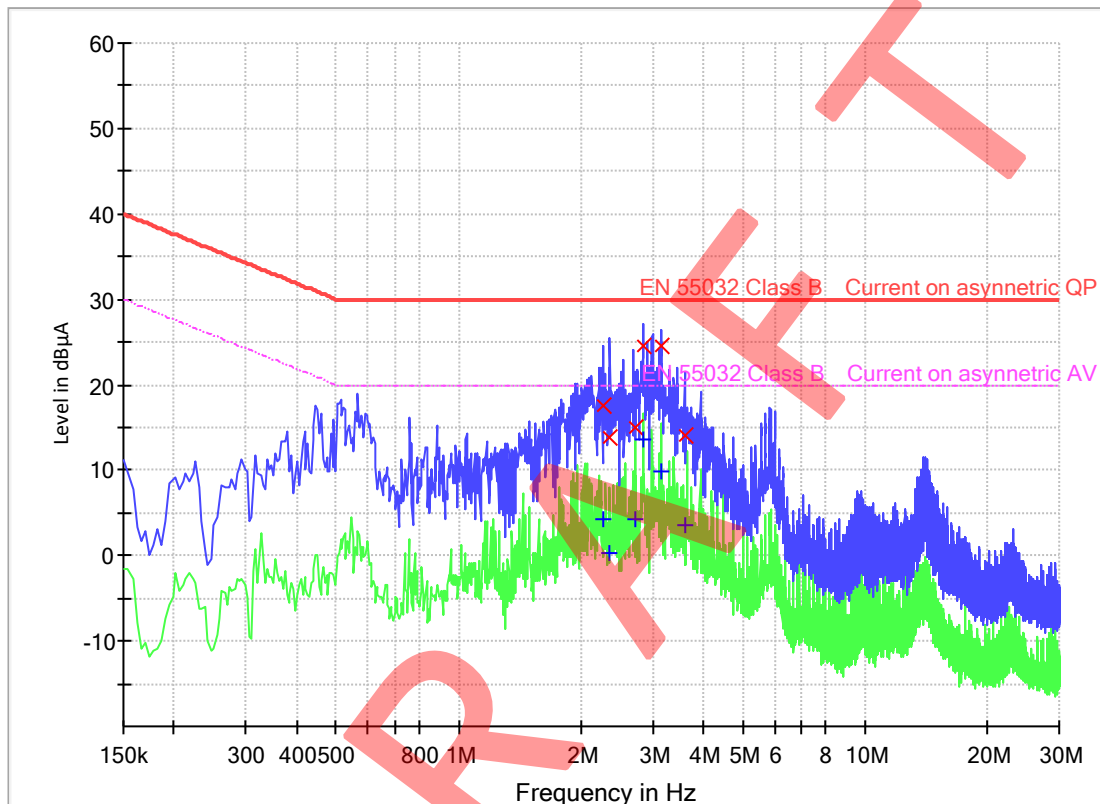
Worst Case Operating Mode: DVB-T(506MHz)

Port: DVB-T Antenna

Panel Model: HF500QUB

Test Data

Asymmetric Mode Conducted Emissions Pursuant to EN55032: Emissions Requirement



Limit and Margin QP

Frequency (MHz)	QuasiPeak (dBuA)	Bandwidth (kHz)	Corr. (dB)	Margin (dB)	Limit (dBuA)
2.268000	17.5	9.000	0.2	12.5	30.0
2.360000	13.9	9.000	0.2	16.1	30.0
2.716000	15.0	9.000	0.2	15.0	30.0
2.848000	24.6	9.000	0.2	5.4	30.0
3.152000	24.7	9.000	0.2	5.3	30.0
3.592000	14.0	9.000	0.3	16.0	30.0

Limit and Margin AV

Frequency (MHz)	Average (dBuA)	Bandwidth (kHz)	Corr. (dB)	Margin (dB)	Limit (dBuA)
2.268000	4.3	9.000	0.2	15.7	20.0
2.360000	0.2	9.000	0.2	19.8	20.0
2.716000	4.2	9.000	0.2	15.8	20.0
2.848000	13.6	9.000	0.2	6.4	20.0
3.152000	10.0	9.000	0.2	10.0	20.0
3.592000	3.5	9.000	0.3	16.5	20.0

Remark:

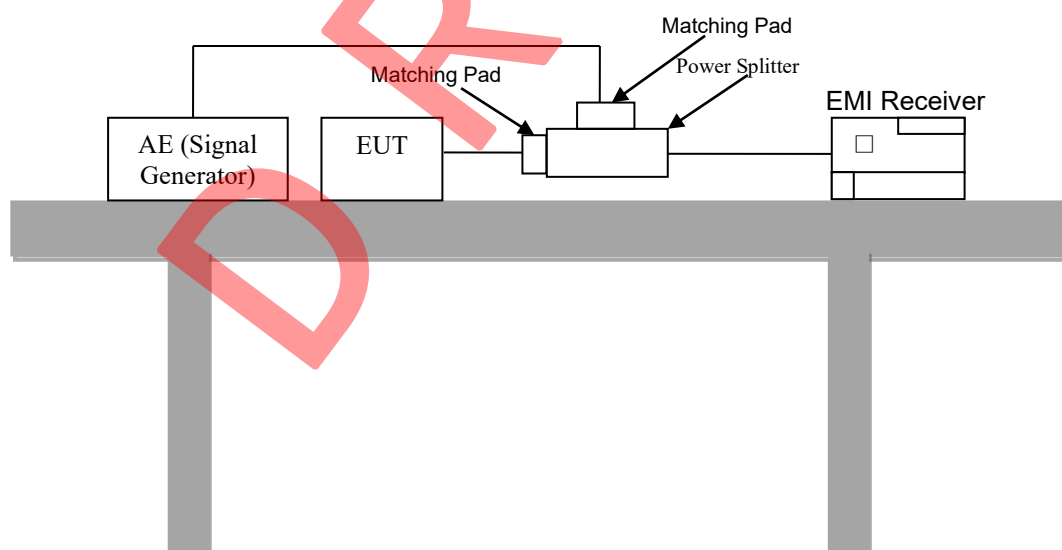
1. Corr. Factor (dB) = Current Probe Factor (dB) + Cable Loss (dB)
2. Margin (dB) = Limit (dBuV) – QuasiPeak/Average (dBuV)

Antenna Terminal Disturbance Voltage
Pursuant to EN55032: Emissions Requirement

Used Test Equipment

Equip No.	Description	Manufacturer	Model No.	Cal. Date	Due Date
SZ185-02	EMI Receiver	R & S	ESCI	2022-07-08	2023-07-08
SZ070-15	Matching Pad	R&S	RAM	2022-05-16	2023-05-16
SZ067-03	Power Splitter	R&S	RVZ	2021-12-21	2022-12-21
				2022-12-19	2023-12-19

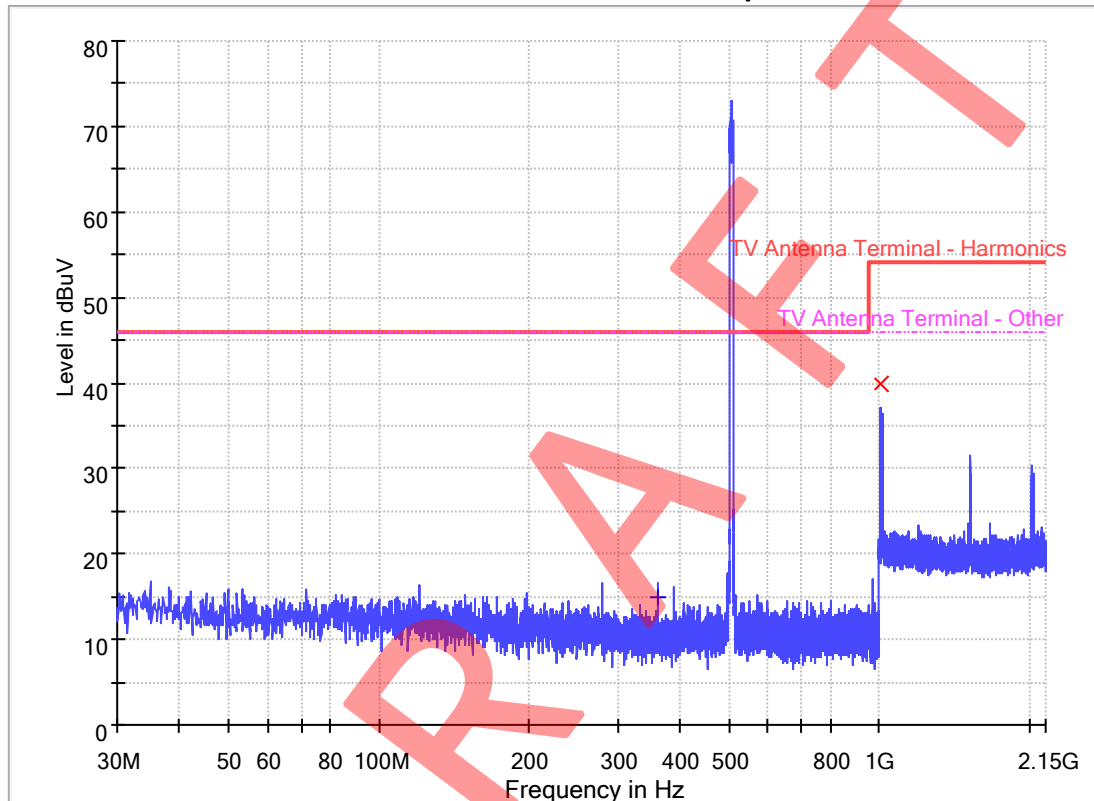
- Notes:1. Peak and average detector quick scan are showed on the graph and final quasi-peak and average detector data are measured, the worst-case is recorded in the following graph and table.
2. Frequency range scanned: 30MHz to 2.15GHz.
3. Only emissions significantly above equipment noise floor is reported.

Test Setup

Reference ground plane

Test Data

Antenna Terminal Disturbance Voltage Pursuant to EN 55032: Emissions Requirement



Limit and Margin

Frequency (MHz)	MaxPeak (dBuV)	QuasiPeak (dB μ V)	Corr. (dB)	Margin (dB)	Limit (dBuV)
1010.781250	39.8	---	9.5	14.2	54.0

Remark:

1. Corr. Factor (dB) = Matching Pad factor (dB) + Power Splitter factor (dB) + Cable Loss (dB)
2. Margin (dB) = Limit (dBuV) – QuasiPeak/Average (dBuV)
3. --- denotes the not applicable

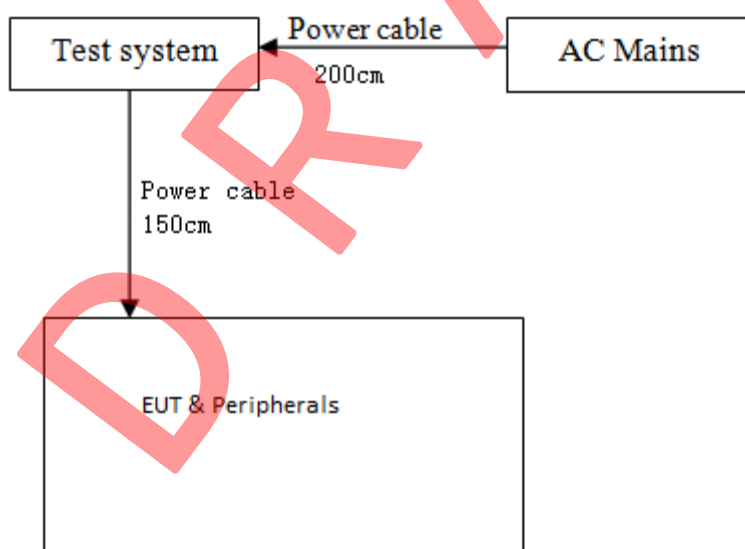
Note: The measured RF signal at FM radio or TV signal carrier frequency and sidebands were not the disturbance of equipment under test.

EN IEC 61000-3-2
Harmonics

Used Test Equipment

Equip No.	Description	Manufacturer	Model No.	Cal. Date	Due Date
SZ064-01	Compliance Test System	California Instruments	5001iX-CTS-400	2023-12-12	2024-12-12
SZ064-01-01	Power Analyzer and Conditioning System	California Instruments	PACS-1	2023-12-12	2024-12-12

Notes: 1. The test result consisting of worst-case was attached in the following pages.
2. Uncertainty: 0.03% at a level of confidence of 95%.

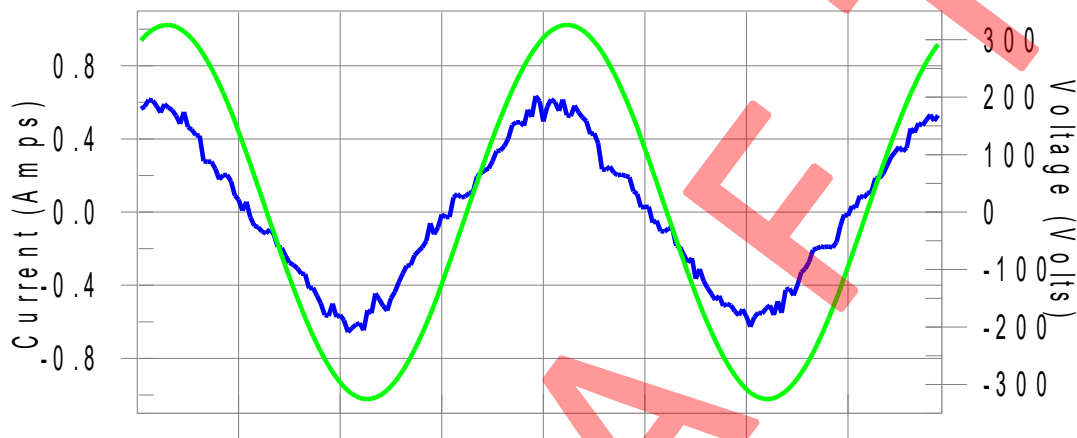
Test Setup Diagram

Harmonics – Class-D per Ed. 4.0 (Run time)

Test Result: Pass

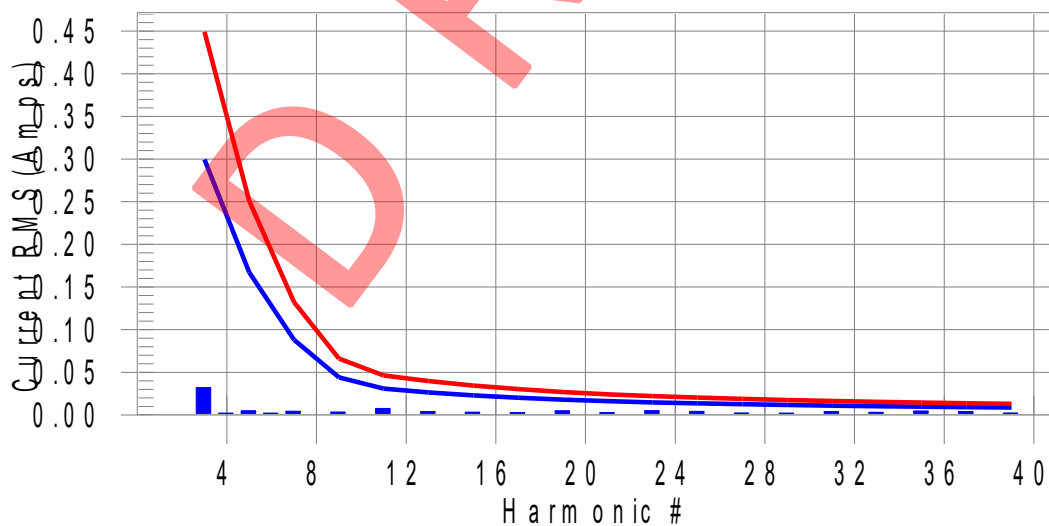
Source qualification: Normal

Current & voltage waveforms



Harmonics and Class D limit line

European Limits



Test result: Pass Worst harmonics H23-29.9% of 150% limit, H23-34.4% of 100% limit

Model: MUDV50B0-90106-T2
Worst Case Operating Mode: DVB-T

Intertek Report No.: 260508031SZN-001

Test Result: Pass Source qualification: Normal
THC(A): 0.037 I-THD(%): 9.4 POHC(A): 0.012 POHC Limit(A): 0.038

Highest parameter values during test:

V_RMS (Volts):	230.34	Frequency(Hz):	50.00
I_Peak (Amps):	0.802	I_RMS (Amps):	0.408
I_Fund (Amps):	0.395	Crest Factor:	2.486
Power (Watts):	88.1	Power Factor:	0.962

Harm#	Harms(avg)	100%Limit	%of Limit	Harms(max)	150%Limit	%of Limit	Status
2	0.002	0.000	N/A	0.002	0.000	N/A	Pass
3	0.032	0.300	10.7	0.037	0.449	8.2	Pass
4	0.002	0.000	N/A	0.003	0.000	N/A	Pass
5	0.005	0.167	N/A	0.006	0.251	N/A	Pass
6	0.002	0.000	N/A	0.002	0.000	N/A	Pass
7	0.004	0.088	N/A	0.006	0.132	N/A	Pass
8	0.002	0.000	N/A	0.002	0.000	N/A	Pass
9	0.004	0.044	N/A	0.008	0.066	N/A	Pass
10	0.001	0.000	N/A	0.002	0.000	N/A	Pass
11	0.008	0.031	24.8	0.009	0.046	19.7	Pass
12	0.001	0.000	N/A	0.002	0.000	N/A	Pass
13	0.004	0.026	N/A	0.005	0.040	N/A	Pass
14	0.001	0.000	N/A	0.002	0.000	N/A	Pass
15	0.003	0.023	N/A	0.006	0.034	N/A	Pass
16	0.002	0.000	N/A	0.002	0.000	N/A	Pass
17	0.003	0.020	N/A	0.004	0.030	N/A	Pass
18	0.001	0.000	N/A	0.002	0.000	N/A	Pass
19	0.005	0.018	N/A	0.007	0.027	N/A	Pass
20	0.001	0.000	N/A	0.002	0.000	N/A	Pass
21	0.003	0.016	N/A	0.005	0.024	N/A	Pass
22	0.002	0.000	N/A	0.002	0.000	N/A	Pass
23	0.005	0.015	34.4	0.007	0.022	29.9	Pass
24	0.002	0.000	N/A	0.002	0.000	N/A	Pass
25	0.004	0.014	N/A	0.005	0.020	N/A	Pass
26	0.002	0.000	N/A	0.002	0.000	N/A	Pass
27	0.002	0.013	N/A	0.003	0.019	N/A	Pass
28	0.002	0.000	N/A	0.002	0.000	N/A	Pass
29	0.002	0.012	N/A	0.003	0.018	N/A	Pass
30	0.002	0.000	N/A	0.002	0.000	N/A	Pass
31	0.004	0.011	N/A	0.005	0.016	N/A	Pass
32	0.002	0.000	N/A	0.002	0.000	N/A	Pass
33	0.003	0.010	N/A	0.005	0.015	N/A	Pass
34	0.002	0.000	N/A	0.002	0.000	N/A	Pass
35	0.005	0.010	N/A	0.006	0.015	N/A	Pass
36	0.002	0.000	N/A	0.002	0.000	N/A	Pass
37	0.004	0.009	N/A	0.005	0.014	N/A	Pass
38	0.002	0.000	N/A	0.002	0.000	N/A	Pass
39	0.002	0.009	N/A	0.003	0.013	N/A	Pass
40	0.001	0.000	N/A	0.002	0.000	N/A	Pass

Note: Dynamic limits were applied for this test. The highest harmonics values in the above table may not occur at the same window as the maximum harmonics/limit ratio.

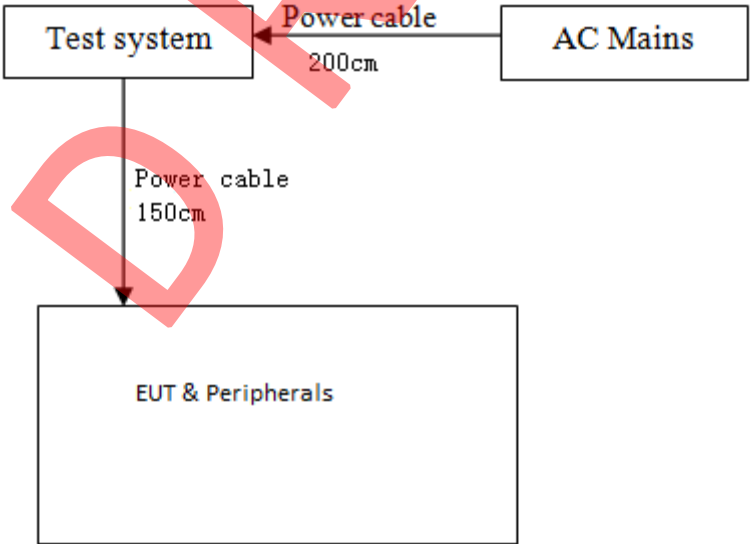
Voltage Fluctuations
Pursuant to EN 61000-3-3

Used Test Equipment

Equip No.	Description	Manufacturer	Model No.	Cal. Date	Due Date
SZ064-01	Compliance Test System	California Instruments	5001iX-CTS-400	2023-12-12	2024-12-12
SZ064-01-01	Power Analyzer and Conditioning System	California Instruments	PACS-1	2023-12-12	2024-12-12

- Notes:
- 1. The test result consisting of worst-case was attached in the following pages.
 - 2. Uncertainty: 0.25% at a level of confidence of 95%.

Test Setup Diagram

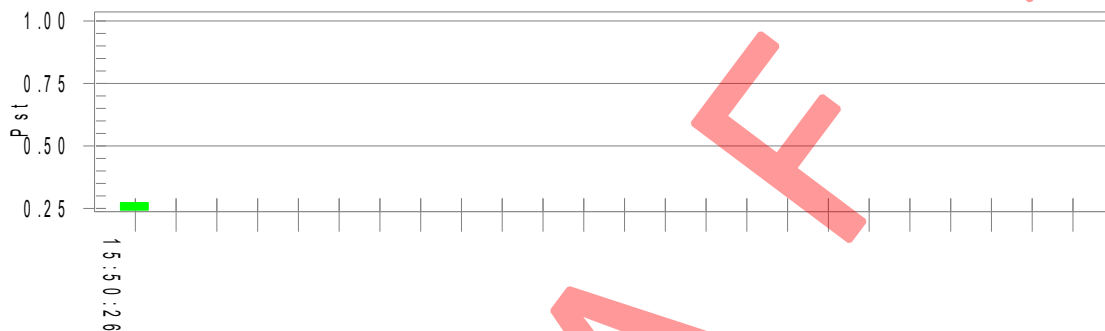


Flicker Test Summary per EN/IEC61000-3-3 Ed. 3.0 (2013) (Run time)

Test Result: Pass

Status: Test Completed

Pst_i and limit line



Parameter values recorded during the test:

Vrms at the end of test (Volt):	230.27		
Highest dt (%):	0	Test limit (%):	
T-max (mS):	0.00	Test limit (mS):	500.0 Pass
Highest dc (%):	0.00	Test limit (%):	3.30 Pass
Highest dmax (%):	0.00	Test limit (%):	4.00 Pass
Highest Pst (10 min. period):	0.273	Test limit:	1.000 Pass

Electrostatic Discharge
Pursuant to EN 61000-4-2

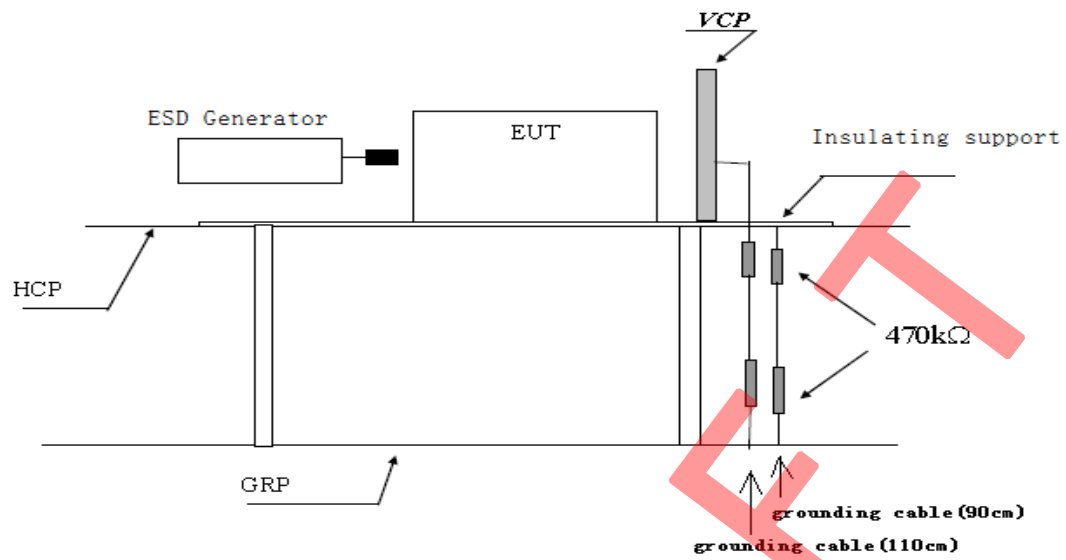
Test Summary (Pursuant to EN 55035)

Port:	Enclosure
Basic Standard:	EN 61000-4-2
Required Performance Criterion:	B
Limit:	±8.0kV (Air Discharge)
	±4.0kV (Contact Discharge)
	±4.0kV (Indirect Contact Discharge)
Temperature:	23.9°C
Relative Humidity:	51.1%
Test Mode:	ATV, DVB-T/T2, DVB-C, AV In, USB Play, HDMI IN, HDMI(ARC), IPTV
Test Setup:	Table-top
Test of Post-Installation:	N/A
Time Between Each Discharge:	1 second

Used Test Equipment

Equip No.	Description	Manufacturer	Model No.	Cal. Date	Due Date
SZ189-05	ESD Simulator	TESEQ	NSG 437	2024-06-07	2025-06-07

Test Setup Diagram



Test set-up of electrostatic discharge

Test Results**EN 61000-4-2
Electrostatic Discharge**

Discharge Type	No. of Discharge	Applied Voltage	Result (Pursuant to EN55035, Criterion B)
Contact Discharge	20	$\pm 4\text{kV}$	Pass
Air Discharge	20	$\pm 2.0, \pm 4.0,$ $\pm 8.0\text{kV}$	Pass
Indirect HCP Discharge	20	$\pm 4\text{kV}$	Pass
Indirect VCP Discharge	20	$\pm 4\text{kV}$	Pass

☒ Additional Information

- ☐ No observable change
- ☐ EUT stopped operation and could not be reset by itself at __V of ESD.
- ☐ EUT was in abnormal operation:
 – Operation mode was changed from ____ to ____ at ____V, ____ of ESD.
- ☒ In all modes, the screen flash during Contact Discharge $\pm 4\text{kV}$ test, but it can be resumed by itself after test

Radiated Immunity
Pursuant to EN 61000-4-3

Test Summary (Pursuant to EN 55035)

Basic Standard:	EN 61000-4-3
Port:	Enclosure
Required Performance Criterion:	A
Limit:	3.0V/m (rms) (see note)
Test Modulation:	1kHz, 80% AM
Frequency:	80MHz to 1000MHz, 1800MHz, 2600MHz, 3500MHz, 5000MHz
Dwell Time:	1s
Frequency Step:	1%
Temperature:	23.2°C
Relative Humidity:	53.3%
Test Facility:	Full Anechoic Chamber
Antenna Polarization:	Horizontal and Vertical
Test Distance:	3 meters
Test Mode:	ATV, DVB-T/T2, DVB-C, AV In, USB Play, HDMI IN, HDMI(ARC), IPTV
Test Setup:	Table-top

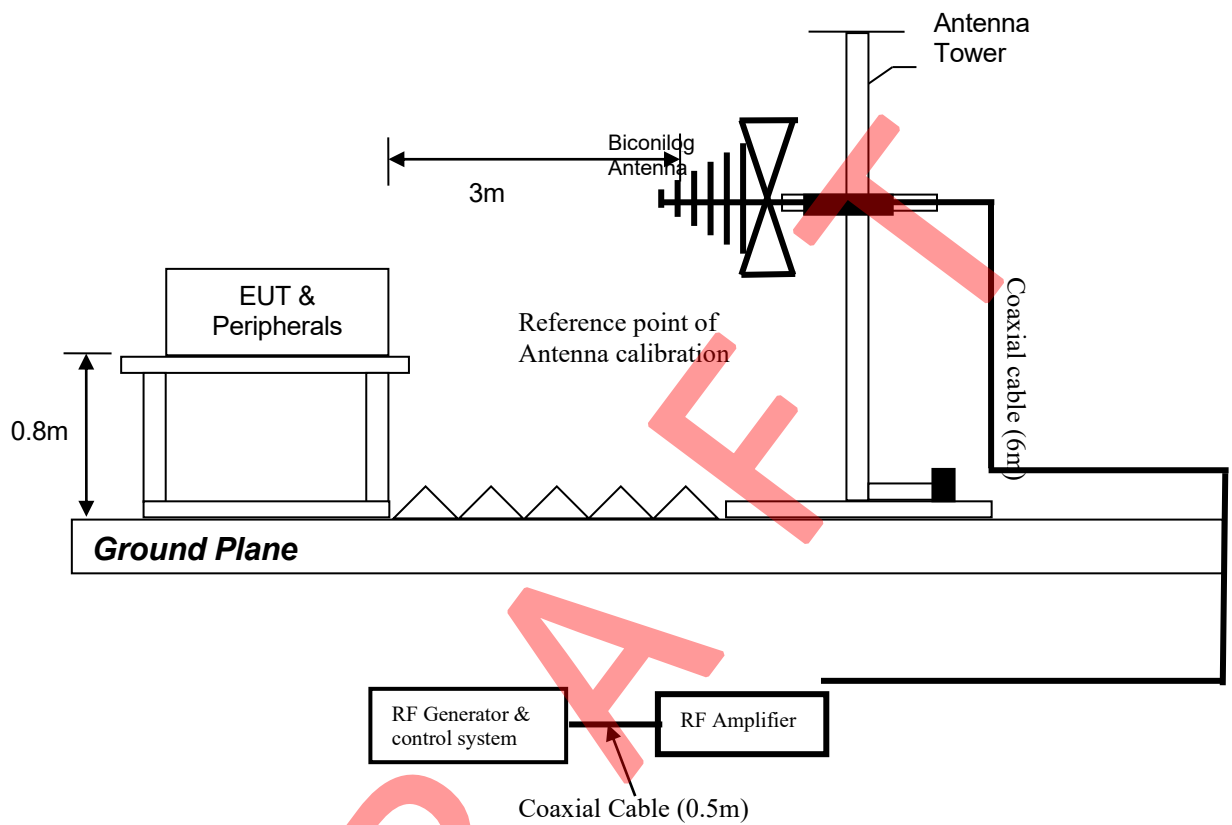
Note: For Broadcast reception function

1. The disturbance level is reduced to 1 V for in-band frequencies.
2. The tuned channel $\pm 0,5$ MHz (lower edge frequency - 0,5 MHz up to the upper edge frequency + 0,5 MHz of the tuned channel) is excluded from testing. Except for DVB-C, the tuned channel $\pm 0,5$ MHz (lower edge frequency - 0,5 MHz up to the upper edge frequency + 0,5 MHz of the tuned channel) is excluded from testing. For DVB-C, the disturbance levels are 3 V/m or 3 V, except in the tuned channel $\pm 0,5$ MHz (lower edge frequency - 0,5 MHz up to the upper edge frequency + 0,5 MHz of the tuned channel), where the disturbance level is 1 V/m.

Used Test Equipment

Equip No.	Description	Manufacturer	Model No.	Cal. Date	Due Date
SZ188-02	Anechoic Chamber	ETS	RFD-F/A-100	2021-12-12	2024-12-12
SZ061-03	Biconilog Antenna	ETS	3142C	2024-07-09	2027-07-09
SZ061-16	Stacked double log. -Per. Antenna	SCHWARZBECK	STLP 9149	2021-11-14	2024-11-14
SZ180-15	Signal Generator	R&S	SMB100A	2023-12-12	2024-12-12
SZ181-01	Amplifier	PRANA	AP32 MT215	2023-12-12	2024-12-12
SZ190-07	RF Amplifier	Milmega	AS0860-75/45	2023-12-12	2024-12-12
SZ089-03	Audio Analyzer	AP	ATS-1A	2023-12-12	2024-12-12

Test Setup Diagram



Test set-up of Immunity to Radiated Electric Fields

Test Results

EN61000-4-3
Radiated Immunity

Frequency (MHz)	Exposed Side	Field Strength V/m (rms)	Result (Pursuant to EN55035, Criterion A)
80 to 1000, 1800, 2600, 3500, 5000	Front	For DVB-C mode: 3V/m(rms), Except the tuned channel ± 0.5 MHz is 1V/m; For others mode: 3.0V/m (rms), Except the in band frequency 1V/m and tuned channel ± 0.5 MHz is excluded from testing	Pass
80 to 1000, 1800, 2600, 3500, 5000	Left		Pass
80 to 1000, 1800, 2600, 3500, 5000	Rear		Pass
80 to 1000, 1800, 2600, 3500, 5000	Right		Pass

Note: For Broadcast reception function

1. The disturbance level is reduced to 1 V for in-band frequencies.
2. The tuned channel ± 0.5 MHz (lower edge frequency - 0.5 MHz up to the upper edge frequency + 0.5 MHz of the tuned channel) is excluded from testing. Except for DVB-C, the tuned channel ± 0.5 MHz (lower edge frequency - 0.5 MHz up to the upper edge frequency + 0.5 MHz of the tuned channel) is excluded from testing. For DVB-C, the disturbance levels are 3 V/m or 3 V, except in the tuned channel ± 0.5 MHz (lower edge frequency - 0.5 MHz up to the upper edge frequency + 0.5 MHz of the tuned channel), where the disturbance level is 1 V/m.

☒ Additional Information

☒ No observable change

☐ EUT stopped operation and could / could not be reset by operator at Freq. _____ of Radiated Immunity.

☐ EUT was in abnormal operation:

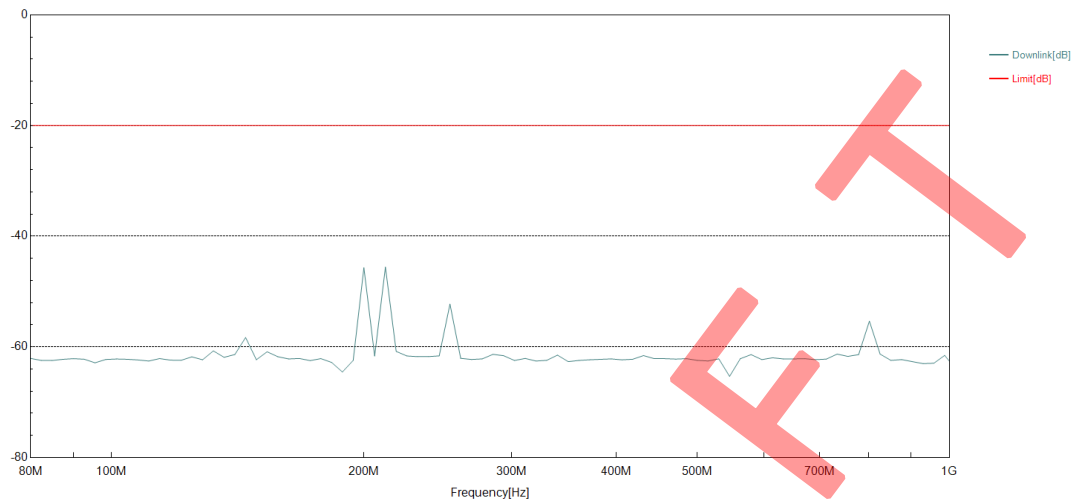
- Operation mode was changed from _____ to _____ at Freq. _____ of Radiated Immunity.

☐ _____

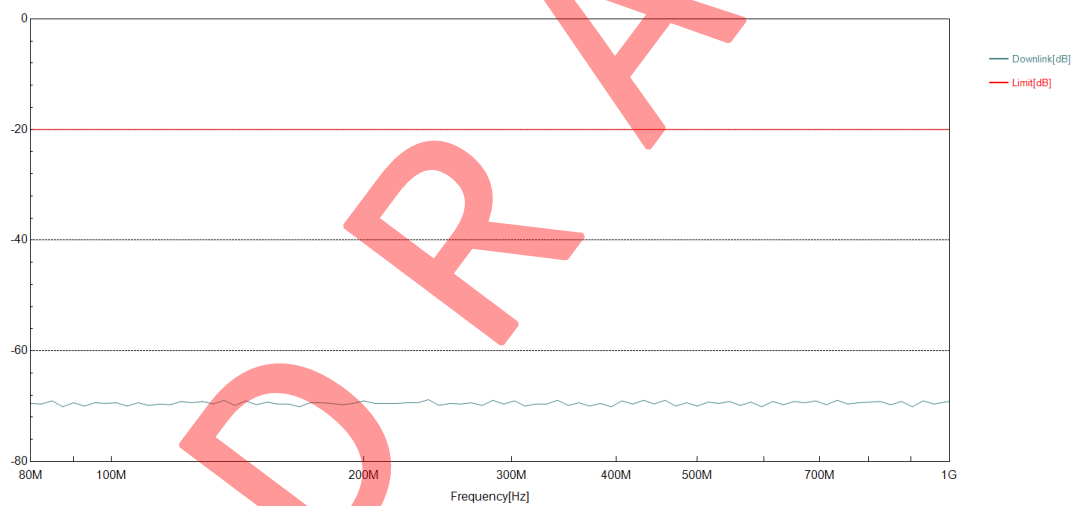
Model: MUDV50B0-90106-T2
Worst Case Operating Mode: DVB-T
Monitor port: Speaker

Intertek Report No.: 260508031SZN-001

Horizontal



Vertical



Mode	Frequency	Port	Polarization	Output Level (dBV)	K/C Level (dBV)	K/C S/N (dB)	Limit(dB)
ATV	3500MHz	Earphone	H	-6.00	-54.28	-48.28	-20.00
			V	-6.00	-63.60	-57.60	-20.00

Electrical Fast Transient / Burst
Pursuant to EN 61000-4-4

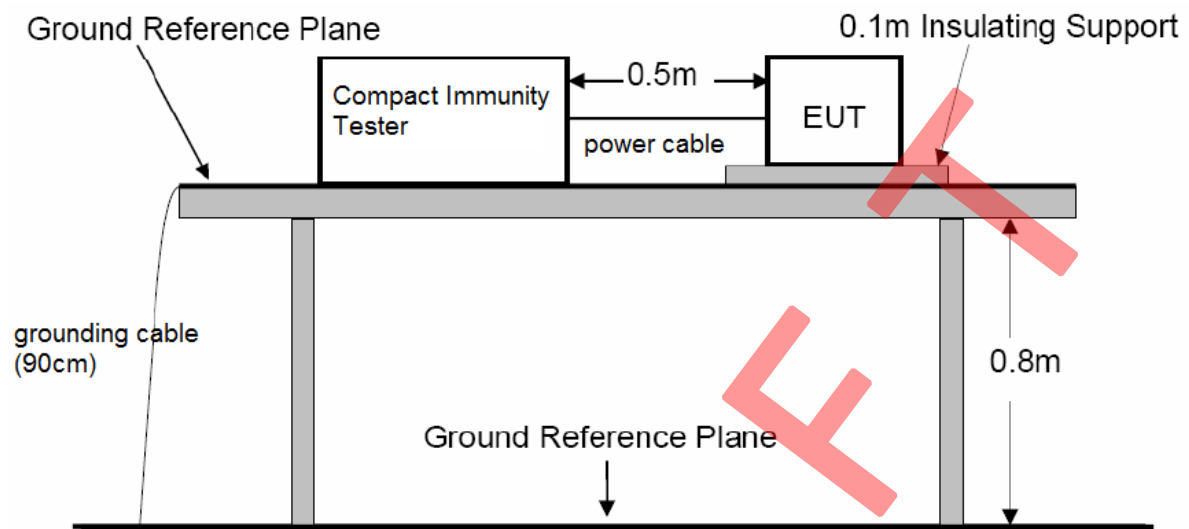
Test Summary (Pursuant to EN 55035)

Basic Standard:	EN 61000-4-4	
Port:	AC Power Lines	Signal Lines
Required Performance Criterion:	B	
Limit:	±1.0kV	±0.5kV
Test Duration:	1 minute	
Temperature:	19.4°C	
Relative Humidity:	49.8%	
Test Mode:	ATV, DVB-T/T2, DVB-C, AV In, USB Play, HDMI IN, HDMI(ARC), IPTV	
Test Setup:	Table-top	
Generator Drive:	Internal	

Used Test Equipment

Equip No.	Description	Manufacturer	Model No.	Cal. Date	Due Date
SZ063-01	Compact Immunity Tester	Haefely	ECOMPACT 4	2021-12-26 2022-12-26	2022-12-26 2023-12-26
SZ063-01-01	Capacitive Coupling Clamp	Haefely	IP4A	2021-12-28 2022-12-27	2022-12-28 2023-12-27

Test Setup Diagram



Test set-up of immunity to electrical fast transient bursts for power port

Test Results**EN61000-4-4
Electrical Fast Transient / Burst**

Port	Level	Polarity	Result (Pursuant to EN55035, Criterion B)
AC Power Lines	1kV	+	Pass
	1kV	–	Pass
Signal Lines	0.5kV	+	Pass
	0.5kV	–	Pass

☒ Additional Information☒ No observable change☐ EUT stopped operation and could / could not be reset by operator at ____V of Fast Transient.☐ EUT was in abnormal operation:
– Operation mode was changed from ____ to ____ at ____V of Fast Transient.☐ _____

Surge Immunity
Pursuant to EN 61000-4-5

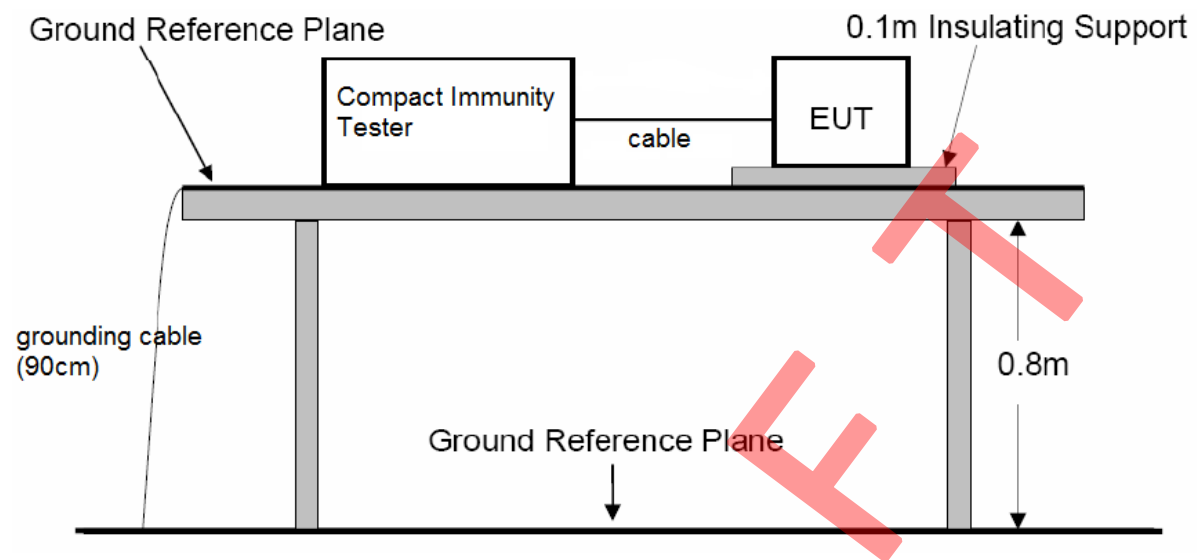
Test Summary (Pursuant to EN 55035)

Basic Standard:	EN 61000-4-5			
Port:	AC Power Lines			Shield
	Phase and Neutral	Phase and Earth	Neutral and Earth	Shield to ground
Limit:	5 Positive and 5 Negative Surges			
	±1kV	±2kV	±2kV	±0.5kV
Generator Impedance:	2ohm	12ohm	12ohm	2ohm
Required Performance Criterion:	B			
Repetition Rate:	1 minute			
Test Mode:	ATV, DVB-T/T2, DVB-C, AV In, USB Play, HDMI IN, HDMI(ARC), IPTV			
Test Setup:	Table-top			
Surge Generator Trigger:	Internal			
Installation Condition:	Class 3: Electrical environment where cables run in parallel.			
Phase Angle:	90°, 270°			

Used Test Equipment

Equip No.	Description	Manufacturer	Model No.	Cal. Date	Due Date
SZ063-01	Compact Immunity Tester	Haefely	ECOMPACT 4	2021-12-26 2022-12-26	2022-12-26 2023-12-26

Test Setup Diagram



Test set-up of Surge Immunity for Power port

Test Results**EN61000-4-5
Surge Immunity**

Level		Result (Pursuant to EN 55035, Criterion B)
Between Phase and Neutral:	$\pm 1\text{kV}$	Pass
Between Phase and Earth:	$\pm 2\text{kV}$	N/A
Between Neutral and Earth:	$\pm 2\text{kV}$	N/A
Between Shield and Earth:	$\pm 0.5\text{kV}$	Pass

☒ Additional Information☒ No observable change☐ EUT stopped operation and could / could not be reset by operator at ____V of Surge.☐ EUT was in abnormal operation:
– Operation mode was changed from ____ to ____ at ____V of Surge.☐

Injected Current (0.15MHz to 80MHz)
Pursuant to EN 61000-4-6

Test Summary (Pursuant to EN 55035)

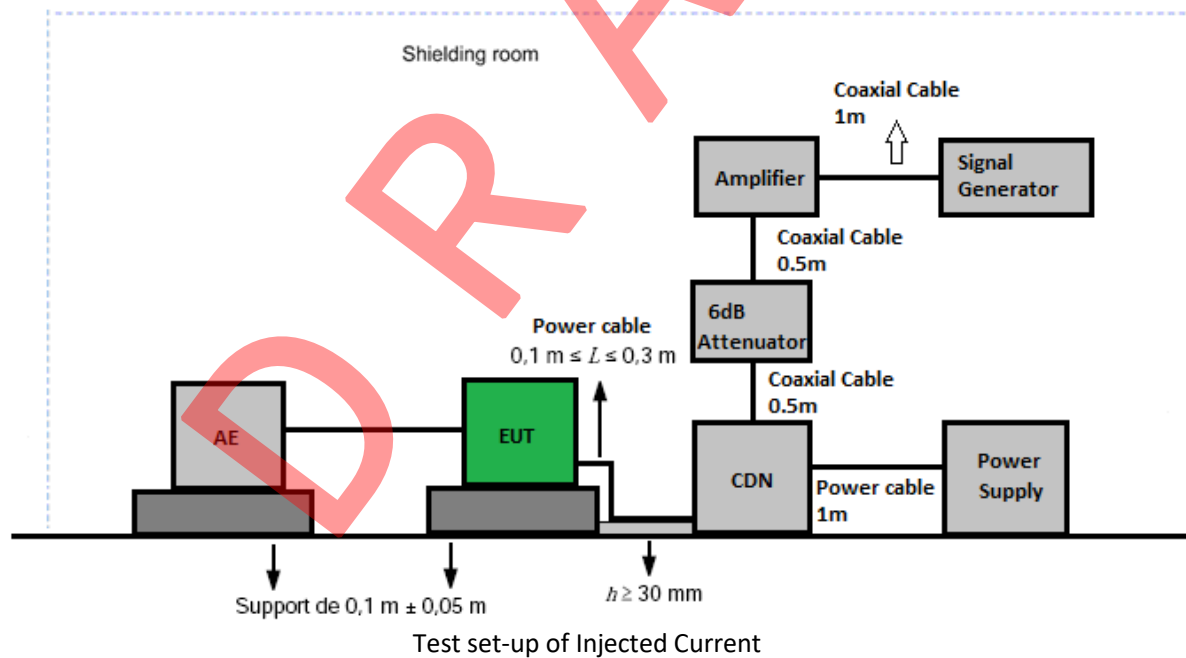
Basic Standard:	EN 61000-4-6	
Port:	AC Power Lines, DC Power Lines, Signal Lines and Control Lines	
Required Performance Criterion:	A	
Level:	0.15MHz-10MHz	3V (rms)
	10MHz-30MHz	3V (rms) to 1V (rms)
	30MHz-80MHz	1V (rms)
Test Modulation:	1kHz, 80% AM	
Frequency:	0.15MHz to 80MHz	
Dwell Time:	1s	
Frequency Step:	1%	
Temperature:	24.0°C	
Relative Humidity:	58.2%	
Coupling Factor of CDN:	-1.0dB ~ -1.7dB	
Test Mode:	ATV, DVB-T/T2, DVB-C, AV In, USB Play, HDMI IN, HDMI(ARC), IPTV	
Test Setup:	Table-top	

Note: For Broadcast reception function

1. The disturbance level is reduced to 1 V for in-band frequencies.
2. The tuned channel $\pm 0,5$ MHz (lower edge frequency - 0,5 MHz up to the upper edge frequency + 0,5 MHz of the tuned channel) is excluded from testing. Except for DVB-C, the tuned channel $\pm 0,5$ MHz (lower edge frequency - 0,5 MHz up to the upper edge frequency + 0,5 MHz of the tuned channel) is excluded from testing. For DVB-C, the disturbance levels are 3 V/m or 3 V, except in the tuned channel $\pm 0,5$ MHz (lower edge frequency - 0,5 MHz up to the upper edge frequency + 0,5 MHz of the tuned channel), where the disturbance level is 1 V/m.

Used Test Equipment

Equip No.	Description	Manufacturer	Model No.	Cal. Date	Due Date
SZ180-02	Signal Generator	Aeroflex	2023A	2021-12-28 2022-12-19	2022-12-28 2023-12-19
SZ181-03	Amplifier	AR-WORLDWIDE	75A250	2021-12-28 2022-12-19	2022-12-28 2023-12-19
SZ181-03-01	Attenuator	AR-WORLDWIDE	6dB/50FH-006-100	2021-12-28 2022-12-19	2022-12-28 2023-12-19
SZ183-01	RF CURRENT-INJECTION CLAMP	LUTHI	EM101	2021-12-27 2022-12-27	2022-12-27 2023-12-27
SZ184-01	Coupling-Decoupling Network	LUTHI	CDN L-801 M2/M3	2021-12-27 2022-12-27	2022-12-27 2023-12-27
SZ089-03	Audio Analyzer	AP	ATS-1A	2021-12-28 2022-12-19	2022-12-28 2023-12-19
SZ188-04	Shielding Room	Jiang yin Tian De	5*6*2.9m/5*2.5*2.7m	2020-01-07	2023-01-07

Test Setup Diagram

Test Results

EN61000-4-6

Injected Current (0.15MHz to 80MHz)

Port	Frequency (MHz)	Level (V)	Result (Pursuant to EN 55035, Criterion A)
AC Power Lines	0.15 to 10	3	Pass
	10 to 30	3 to 1	
	30 to 80	1 Except for DVB-C, the tuned channel ± 0.5 MHz is excluded from testing	
Signal Lines	0.15 to 10	3	Pass
	10 to 30	3 to 1	
	30 to 80	1 Except for DVB-C, the tuned channel ± 0.5 MHz is excluded from testing	

Note: For Broadcast reception function

1. The disturbance level is reduced to 1 V for in-band frequencies.
2. The tuned channel ± 0.5 MHz (lower edge frequency - 0.5 MHz up to the upper edge frequency + 0.5 MHz of the tuned channel) is excluded from testing. Except for DVB-C, the tuned channel ± 0.5 MHz (lower edge frequency - 0.5 MHz up to the upper edge frequency + 0.5 MHz of the tuned channel) is excluded from testing. For DVB-C, the disturbance levels are 3 V/m or 3 V, except in the tuned channel ± 0.5 MHz (lower edge frequency - 0.5 MHz up to the upper edge frequency + 0.5 MHz of the tuned channel), where the disturbance level is 1 V/m.

☒ Additional Information

☒ No observable change

☐ EUT stopped operation and could / could not be reset by operator at ____ V of Injected Current.

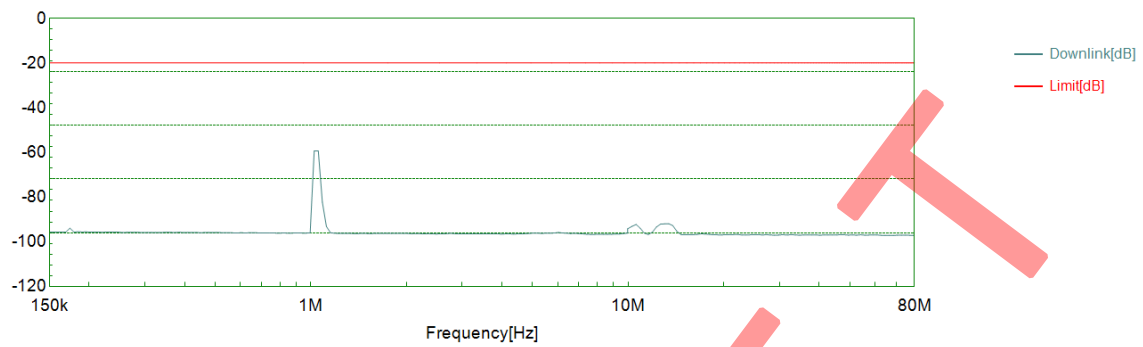
☐ EUT was in abnormal operation:
 – Operation mode was changed from ____ to ____ at ____ V of Injected Current.

☐ _____

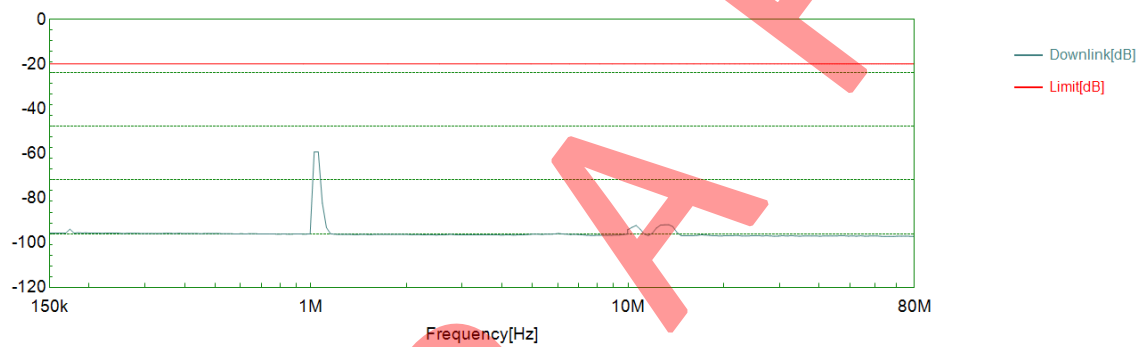
Model: MUDV50B0-90106-T2
Worst Case Operating Mode: DVB-T
Monitor: Speaker

Intertek Report No.: 260508031SZN-001

AC Port:



Signal Port:



Voltage Dips and Interruptions
Pursuant to EN 61000-4-11

Test Summary (Pursuant to EN 55035)

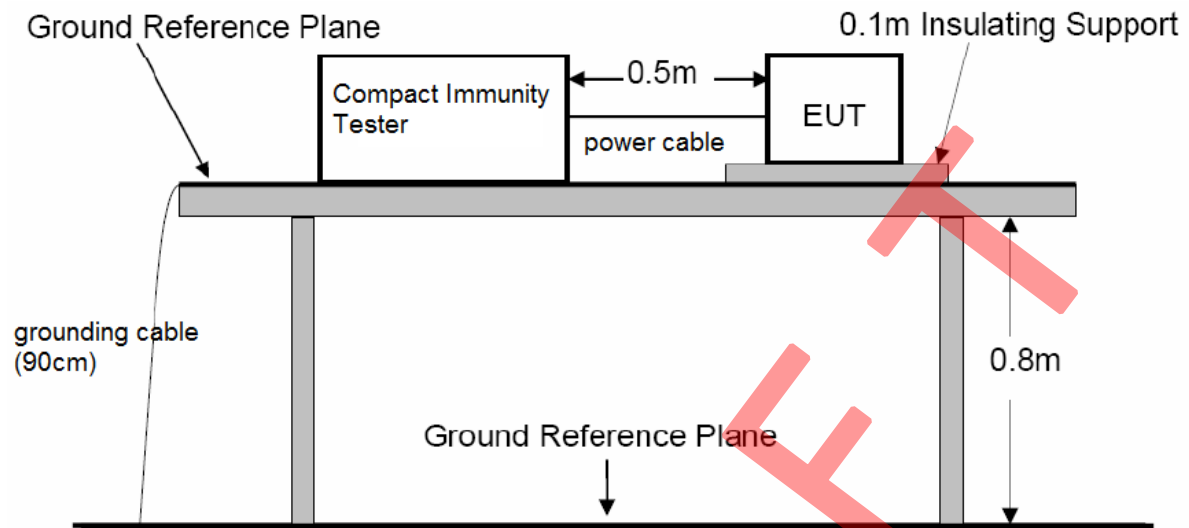
Basic Standard:	EN61000-4-11		
Port:	AC Power Lines		
Limit:	Test Level in %U _T	Duration(s)	Required Performance Criterion
	0	0.01	B
	70	0.5	C
	0	5	C
No. of Dips / Interruptions:	3		
Test Mode:	ATV, DVB-T/T2, DVB-C, AV In, USB Play, HDMI IN, HDMI(ARC), IPTV		
Test Setup:	Table-top		

U_T is the rated voltage for the equipment.

Used Test Equipment

Equip No.	Description	Manufacturer	Model No.	Cal. Date	Due Date
SZ063-01	Compact Immunity Tester	Haefely	ECOMPACT 4	2021-12-26 2022-12-26	2022-12-26 2023-12-26

Test Setup Diagram



Test set-up of Voltage Dips and Interruptions

Test Results**EN61000-4-11
Voltage Dips and Interruptions**

Test Condition		Result (Pursuant to EN 55035, Criterion B)
Test Level in %U _T	Duration(s)	
0	0.01	Pass

Test Condition		Result (Pursuant to EN 55035, Criterion C)
Test Level in %U _T	Duration(s)	
70	0.5	Pass
0	5	Pass

☒ Additional Information☐ No observable change☒ EUT stopped operation and could be reset by operator at test level 0%UT,5S of Interrupt.☐ EUT was in abnormal operation:

– Operation mode was changed from _____ to _____ at test level _____ of Dip. / Interrupt.

☐

Photos of Test Set-up

Please refer to Annex of EUT Photos_260508031SZN.

Photos of EUT

Please refer to Annex of EUT Photos_260508031SZN.

***** End of Report*****

DRAFT